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MIL-M-28787/264B
25 May 1994
SUPERSEDING
MIL-M-28787/264A(EC)
21 July 1983

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FLIP-FLOP, JK, DIGITAL, KEY CODE EMQ

*This specification is approved for use by all Departments and Agencies of the Department of Defense.

*The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic JK flip-flop module.

1.2 Classification. This specification covers two classes of modules. Class I and class II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Code 603A, Bldg. 2917, NAVSURFWARCENDIV, 300 Highway 361, Crane, IN 47522-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

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3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

* 3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.342 failures per million operating hours for class I or 3.344 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

* 3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

* 3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be EMQ.

3.6.2 Size. The module size shall be 1A.

3.6.3 Weight. The module shall have a maximum weight of 35 grams.

3.6.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/264-1 for class I and M28787/264-2 for class II. No marking shall interfere with heat

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dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.8 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 3 to 18 V ±5%.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

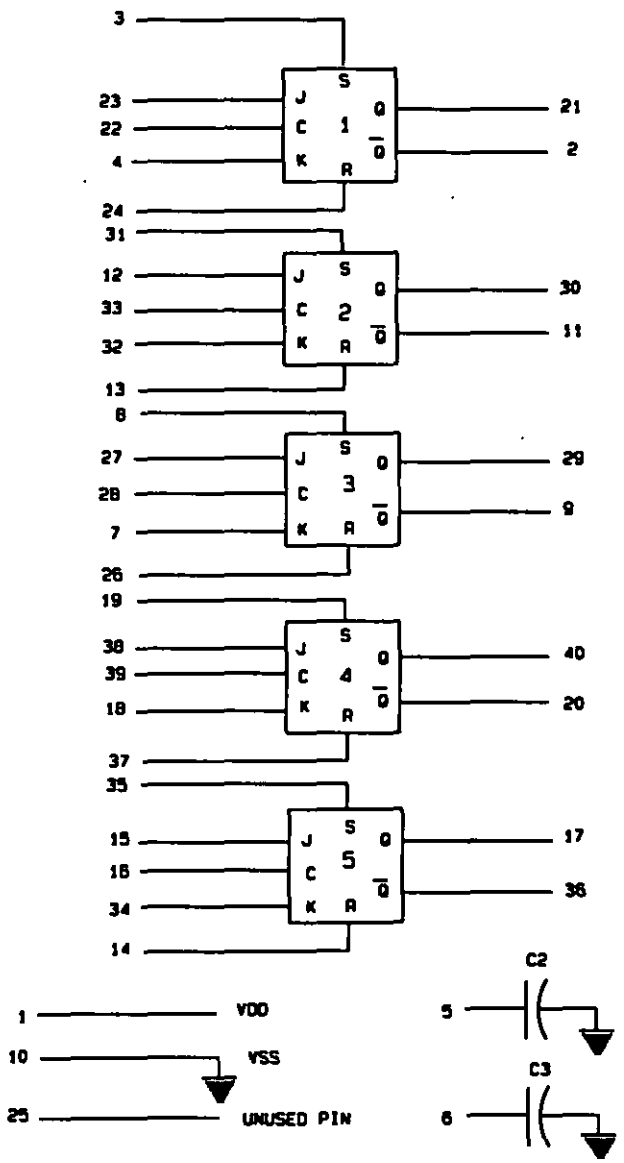
* 4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

* 4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

* 4.4.2 Order of testing. Test conditions A through LL shall be performed in any order with the following exceptions. Test conditions A, B, and C are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, and C where applicable, followed by the remaining tests in the sequence.

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NOTES:
1. J,K = data inputs.
2. C = clock input.
3. S = set input.
4. R = reset input.
5. Q = data output.
6. $\overline{Q}$ = complement of Q data output.

C2 = C3 = 220 pF ± 20 percent.

FIGURE 1. EMQ module function diagram, JK flip-flop.

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* TABLE I. Module electrical requirements.

Test cond	Parameter	Ref test table	Test type 1/ - 3/	25° ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0° ± 5°C		60° ± 5°C		-55° ± 5°C		85° ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1,A2	Isolation Initial End-of-life	III III	S S	10 1	INF INF	10 1	INF INF	10 1	INF INF	10 1	INF INF	10 1	INF INF	MOhm MOhm
B1-B25	Negative input clamp diode voltage	IV	S	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C25	Positive input clamp diode voltage	V	S	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D7	Functional	VI	X	<----------				See table VI					---------->	
E1-E10	Output voltage (low level), (IOL = 0)	VII	X	0	50	0	50	0	50	0	50	0	50	mV
F1-F10	Output voltage (high level), (IOH = 0)	VII	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
G1-G10	Output voltage (low level), VDD = 5V	VIII	X	0	400	0	400	0	400	0	400	0	400	mV
H1-H10	Output voltage (high level), VDD = 5V	IX	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
J1-J10	Output voltage (low level), VDD = 15V	X	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
K1-K10	Output voltage (high level), VDD = 15V	XI	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
L1-L25	Input current (low level)	XII	X	0	-200	---	---	0	-200	---	---	0	-200	nA
M1-M25	Input current (high level)	XIII	X	0	200	---	---	0	200	---	---	0	200	nA
N1	Power supply current (low level)	XIV	X	0	60	0	60	0	590	0	60	0	985	µA
N2	Power supply current (high level)	XIV	X	0	60	0	60	0	590	0	60	0	985	µA
P1-P10	Transition time, tTLH, CL = 50 pF, VDD = 5V	XV	S	11	360	11	360	12	400	8	275	13	425	nS

-See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/ - 3/	25° ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0° ± 5°C		60° ± 5°C		-55° ± 5°C		85° ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
P11-P20	Transition time, tTLH, CL = 150 pF, VDD = 5V	XV	Q	25	840	25	840	28	930	19	640	29	967	nS
Q1-Q10	Transition time, tTHL, CL = 50 pF, VDD = 5V	XV	S	6	200	6	200	7	220	5	150	7	235	nS
Q11-Q20	Transition time, tTHL, CL = 150 pF, VDD = 5V	XV	Q	20	680	20	680	23	750	16	515	24	802	nS
R1-R10	Propagation delay time, tPLH, (clock to output), CL = 50 pF, VDD = 5V	XVI	S	11	350	11	350	12	385	8	265	12	412	nS
R11-R20	Propagation delay time, tPLH, (clock to output), CL = 150 pF, VDD = 5V	XVI	Q	17	550	17	550	18	610	13	420	20	650	nS
S1-S10	Propagation delay time, tPHL, (clock to output), CL = 50 pF, VDD = 5V	XVI	S	11	350	11	350	12	385	8	265	12	412	nS
S11-S20	Propagation delay time, tPHL, (clock to output), CL = 150 pF, VDD = 5V	XVI	Q	17	550	17	550	18	610	13	420	20	650	nS
T1-T5	Propagation delay time, tPLH, (set to output), CL = 50 pF, VDD = 5V	XVII	S	11	350	11	350	12	385	8	265	12	412	nS
T6-T10	Propagation delay time, tPLH, (reset to output), CL = 50 pF, VDD = 5V	XVII	S	14	450	14	450	15	500	10	340	16	531	nS
T11-T15	Propagation delay time, tPLH, (set to output), CL = 150 pF, VDD = 5V	XVII	Q	17	550	17	550	18	610	13	420	20	650	nS
T16-T20	Propagation delay time, tPLH, (reset to output), CL = 150 pF, VDD = 5V	XVII	Q	20	650	20	650	22	720	15	495	23	787	nS

-See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/ - 3/	25° ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0° ± 5°C		60° ± 5°C		-55° ± 5°C		85° ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
U1-U5	Propagation delay time, tPHL, (set to output), CL = 50 pF, VDD = 5V	XVII	S	11	350	11	350	12	385	8	265	12	412	nS
U6-U10	Propagation delay time, tPHL, (reset to output), CL = 50 pF, VDD = 5V	XVII	S	14	450	14	450	15	500	10	340	16	531	nS
U11-U15	Propagation delay time, tPHL, (set to output), CL = 150 pF, VDD = 5V	XVII	Q	17	550	17	550	18	610	13	420	20	650	nS
U16-U20	Propagation delay time, tPHL, (reset to output), CL = 150 pF, VDD = 5V	XVII	Q	20	650	20	650	22	720	15	495	23	767	nS
V1-V10	Transition time, tTLH, CL = 50 pF, VDD = 10V	XVIII	Q	5	180	5	180	6	200	4	135	6	212	nS
W1-W10	Transition time, tTHL, CL = 50 pF, VDD = 10V	XVIII	Q	3	100	3	100	3	110	2	75	4	120	nS
X1-X10	Propagation delay time, tPLH, (clock to output), CL = 50 pF, VDD = 10V	XIX	Q	5	150	5	150	5	165	3	115	5	177	nS
Y1-Y10	Propagation delay time, tPHL, (clock to output), CL = 50 pF, VDD = 10V	XIX	Q	5	150	5	150	5	165	3	115	5	177	nS
Z1-Z5	Propagation delay time, tPLH, (set to output), CL = 50 pF, VDD = 10V	XX	Q	5	150	5	150	5	165	3	115	5	177	nS
Z6-Z10	Propagation delay time, tPLH, (reset to output), CL = 50 pF, VDD = 10V	XX	Q	6	200	6	200	7	220	5	150	7	235	nS

-See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/ - 3/	25° ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0° ± 5°C		60° ± 5°C		-55° ± 5°C		85° ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
AA1-AA5	Propagation delay time, tPHL, (set to output), CL = 50 pF, VDD = 10V	XX	Q	5	150	5	150	5	165	3	115	5	177	nS
AA6-AA10	Propagation delay time, tPHL, (reset to output), CL = 50 pF, VDD = 10V	XX	Q	6	200	6	200	7	220	5	150	7	235	nS
BB1-BB10	Transition time, tTLH, CL = 50 pF, VDD = 15V	XXI	Q	4	130	4	130	4	145	3	100	5	155	nS
CC1-CC10	Transition time, tTHL, CL = 50 pF, VDD = 15V	XXI	Q	2	80	2	80	3	90	2	60	3	98	nS
DD1-DD10	Propagation delay time, tPLH, (clock to output), CL = 50 pF, VDD = 15V	XXII	Q	3	100	3	100	3	110	2	75	4	120	nS
EE1-EE10	Propagation delay time, tPHL, (clock to output), CL = 50 pF, VDD = 15V	XXII	Q	3	100	3	100	3	110	2	75	4	120	nS
FF1-FF5	Propagation delay time, tPLH, (set to output), CL = 50 pF, VDD = 15V	XXIII	Q	3	100	3	100	3	110	2	75	4	120	nS
FF6-FF10	Propagation delay time, tPLH, (reset to output), CL = 50 pF, VDD = 15V	XXIII	Q	5	150	5	150	5	165	4	115	5	177	nS
GG1-GG5	Propagation delay time, tPHL, (set to output), CL = 50 pF, VDD = 15V	XXIII	Q	3	100	3	100	3	110	2	75	15	120	nS
GG6-GG10	Propagation delay time, tPHL, (reset to output), CL = 50 pF, VDD = 15V	XXIII	Q	5	150	5	150	5	165	4	115	5	177	nS

-See footnotes on next page.

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* TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/ - 3/	25° ± 5°C 4/		Class 1 4/				Class II 4/				Unit
						0° ± 5°C		60° ± 5°C		-55° ± 5°C		85° ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
HH1-HH10	Maximum shift frequency (TA)	XXIV	S	1200	1465	1200	1465	1200	1465	1200	1465	1200	1465	nS
JJ	Power supply filter capacitance	XXV	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
KK1,KK2	C2 and C3 capacitance	XXVI	X	176	264	176	264	176	264	176	264	176	264	pF
LL1-LL26	Input capacitance	XXVII	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

* 4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through LL in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through K and N through KK shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Transient thermal test. For the transient thermal test, use the same test conditions as the maximum class temperature conditions.

4.5.1.2 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

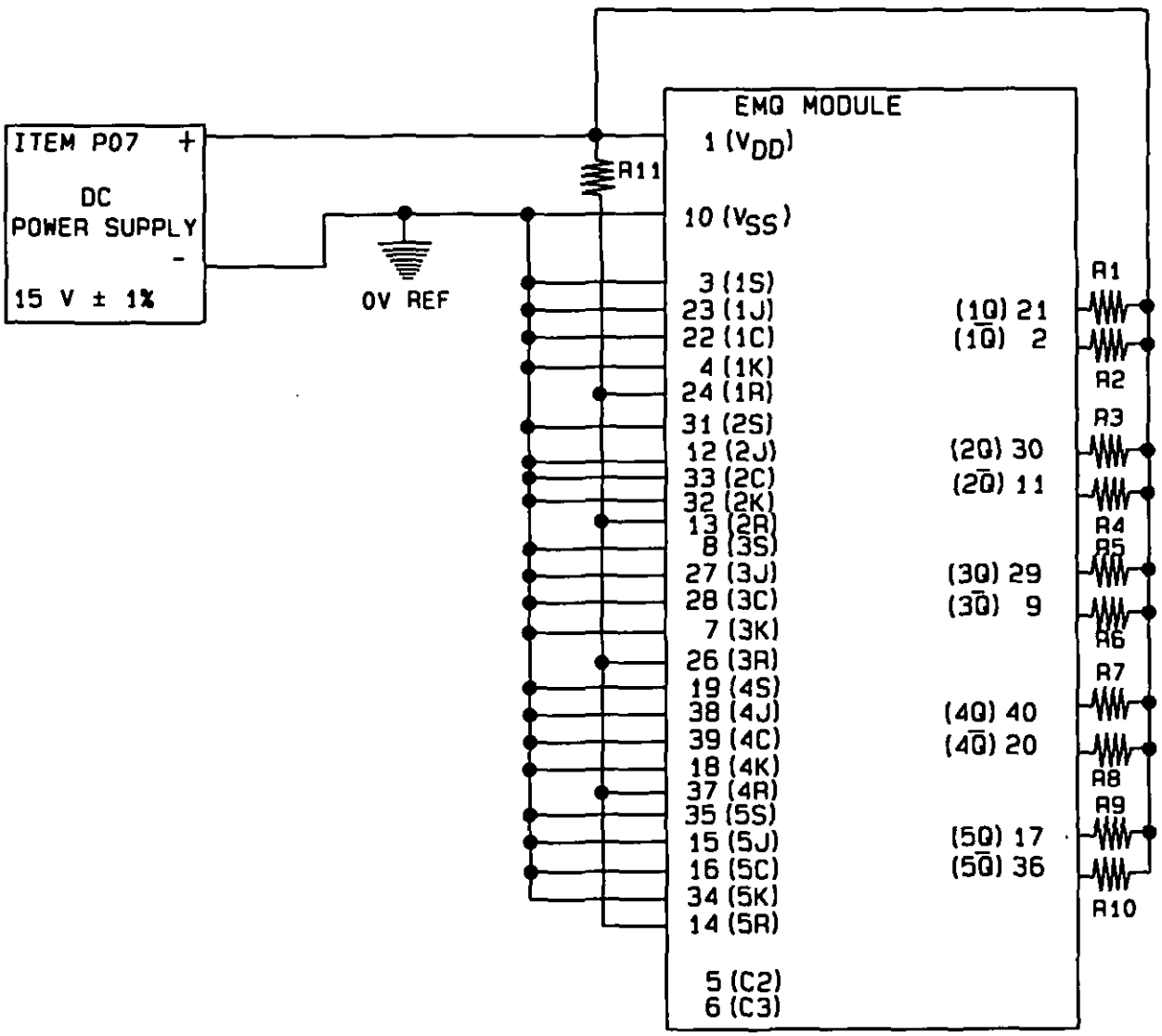
4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition JJ, shall be performed using the initial limits at 25 ± 5°C ambient.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through KK shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25° ± 5°C, electrical tests D through LL shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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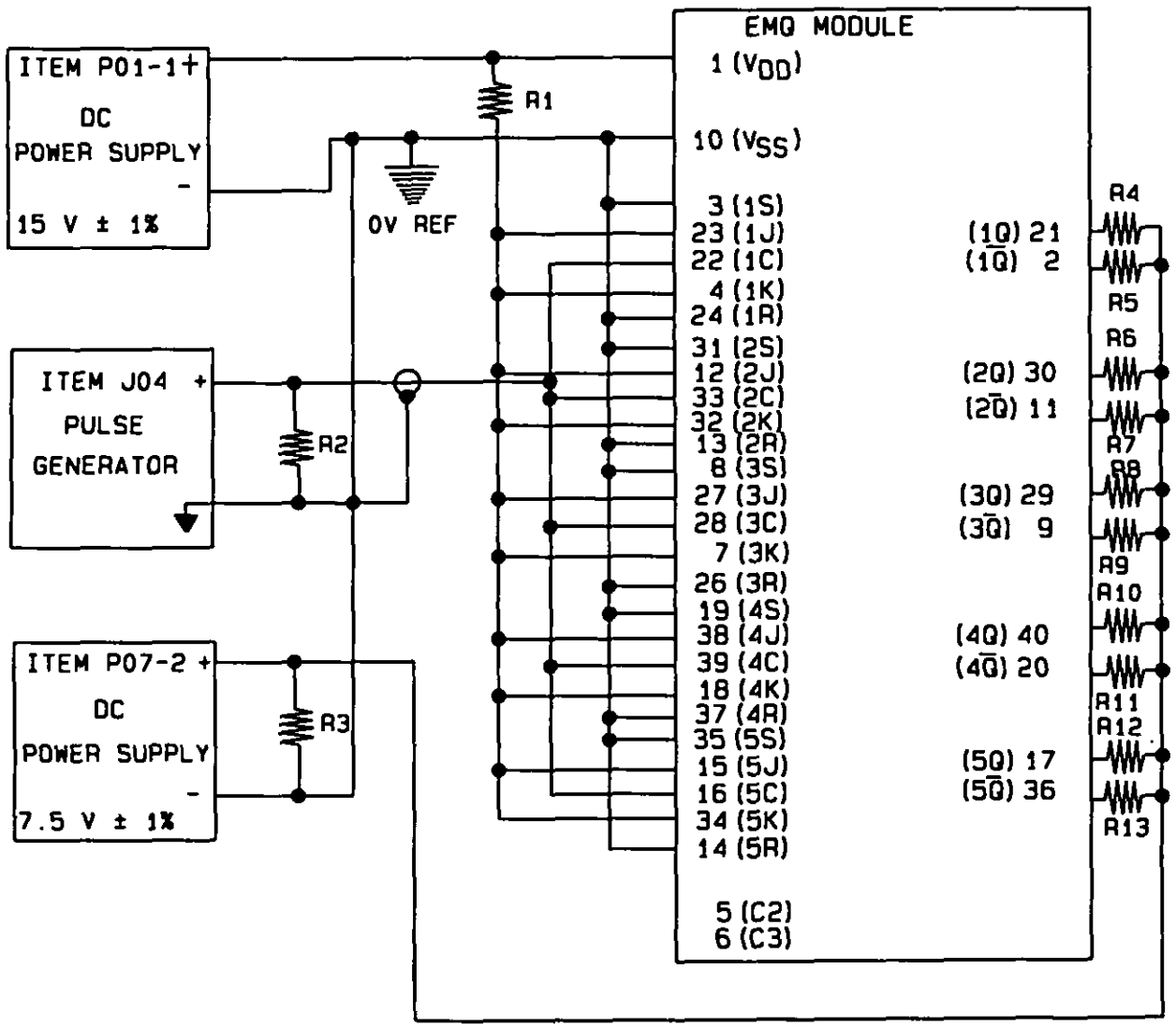


NOTES:

1. R1 = R3 = R5 = R7 = R9 = 6.81 kOhms ± 1 percent, 1/8 W.
2. R2 = R4 = R6 = R8 = R10 = 1.37 kOhms ± 1 percent, 1/8 W.
3. R11 = 1.5 kOhms ± 1 percent, 1/4 W.

* FIGURE 2. Operating temperature and life test circuit.

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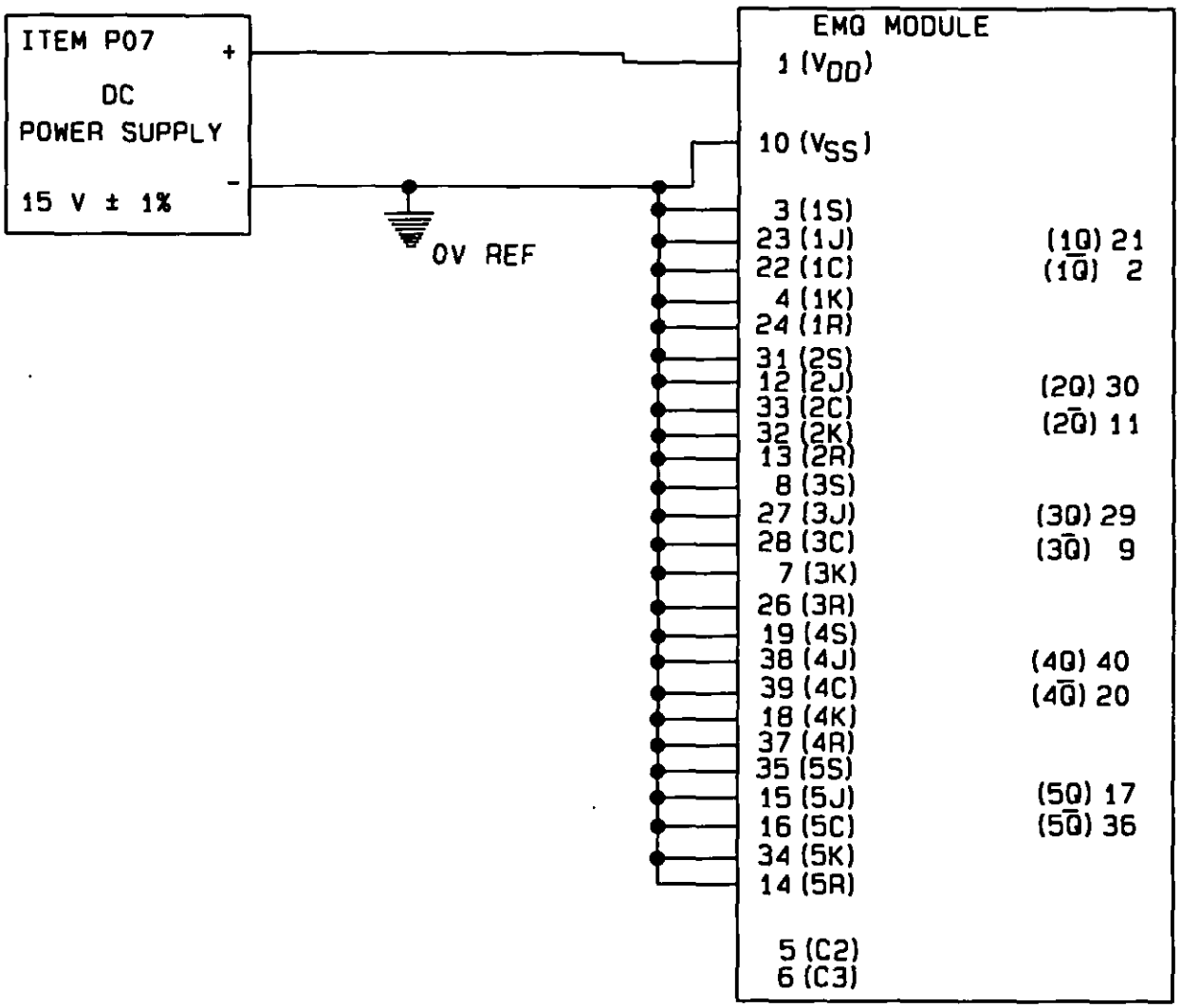


NOTES:

1. R1 = 1.5 kOhms ± 1 percent, 1/4 W.
2. R2 = R3 = 51.1 Ohms ± 5 percent, 5 W.
3. Item J04 setup:
   PA = 15 V ± 1 percent.
   PW = 1.0 µS ± 200 nS.
   Frequency = 500 kHz ± 20 percent.
   tTLH = tTHL = 50 ± 10 nS.
4. R4-R13 = 10 kOhms ± 1 percent, 1/4 W.

* FIGURE 3. Component temperature test circuit.

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* FIGURE 4. Vibration test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

* 6.3 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 85°C end-of-life limits in table I.

	Class I	Class II	Unit
Recommended supply voltage	3 to 18	3 to 18	V dc
Quiescent maximum power dissipation (15 V dc operation) (outputs not loaded)	18	18	mW
Input voltage			
Low level (VDD = 5.0V)	0.0 min,	0.0 min,	V
At -55° to 85°C	1.5 max	1.5 max	V
Low level (VDD = 15.0V)	0 min,	0 min,	V
At -55° to 85°C	4 max	4 max	V
High level (VDD = 5.0V)	3.5 min,	3.5 min,	V
At -55° to 85°C	VDD max	VDD max	V
High level (VDD = 15.0V)	11 min,	11 min,	V
At -55° to 85°C	VDD max	VDD max	V
Input current			
Low level (VIL = 0V, VDD = 15V)	0 min,	0 min,	nA
	-200 max	-200 max	nA
High level (VIH = 15V, VDD = 15V)	0 min,	0 Min,	nA
	200 max	200 max	nA

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	Class I	Class II	Unit
Output voltage			
Low level (VDD = 5.0V)			
At -55° to 85°C	0 min,	0 min,	mV
(IOL = 0)	50 max	50 max	mV
At -55°C	0 min,	0 min,	mV
(IOL = 0.64 mA)	400 max	400 max	mV
At 0° and 25°C	0 min,	0 min,	mV
(IOL = 0.51 mA)	400 max	400 max	mV
At 60°C	0 min,	0 min,	mV
(IOL = 0.44 mA)	400 max	400 max	mV
At 85°C	0 min,	0 min,	mV
(IOL = 0.36 mA)	400 max	400 max	mV
High level (VDD = 5.0V)			
At -55° to 85°C	4.95 min,	4.95 min,	V
(IOH = 0)	5.05 max	5.05 max	V
At -55°C	4.6 min,	4.6 min,	V
(IOH = -0.64 mA)	5.0 max	5.0 max	V
At 0° and 25°C	4.6 min,	4.6 min,	V
(IOH = -0.51 mA)	5.0 max	5.0 max	V
At 60°C	4.6 min,	4.6 min,	V
(IOH = -0.44 mA)	5.0 max	5.0 max	V
At 85°C	4.6 min,	4.6 min,	V
(IOH = -0.36 mA)	5.0 max	5.0 max	V
Low level (VDD = 15.0 V)			
At -55° to 85°C	0.0 min,	0.0 min	V
(IOL = 0)	1.5 max	1.5 max	V
High level (VDD = 15.0V)			
At -55° to 85°C	13.5 min,	13.5 min,	V
(IOH = 0)	15.0 max	15.0 max	V
Transition time			
VDD = 5V			
tTLH (CL = 50pF)	12 min,	13 min,	nS
	400 max	425 max	nS
tTLH (CL = 150pF)	28 min,	29 min,	nS
	930 max	967 max	nS

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	Class I	Class II	Unit
Transition time			
VDD = 5V			
tTHL (CL = 50pF)	7 min,	7 min,	nS
	220 max	235 max	nS
tTHL (CL = 150pF)	23 min,	24 min,	nS
	750 max	802 max	nS
VDD = 10V			
tTLH (CL = 50pF)	6 min,	6 min,	nS
	200 max	212 max	nS
tTHL (CL = 50pF)	3 min,	4 min,	nS
	110 max	120 max	nS
VDD = 15V			
tTLH (CL = 50pF)	5 min,	5 min,	nS
	145 max	155 max	nS
tTHL (CL = 50pF)	3 min,	3 min,	nS
	90 max	96 max	nS
Propagation delay time			
VDD = 5V			
tPLH			
Clock to output			
(CL = 50 pF)	12 min,	12 min,	nS
	385 max	412 max	nS
(CL= 150 pF)	18 min,	20 min,	nS
	610 max	650 max	nS
Set to output			
(CL = 50 pF)	12 min,	12 min,	nS
	385 max	412 max	nS
(CL = 150 pF)	18 min,	20 min,	nS
	610 max	650 max	nS
Reset to output			
(CL = 50 pF)	15 min,	16 min,	nS
	500 max	531 max	nS
(CL = 150 pF)	22 min,	23 min,	nS
Set to output	720 max	767 max	nS
tPHL			
Clock to output			
(CL = 50 pF)	12 min,	12 min,	nS
	385 max	412 max	nS

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	Class I	Class II	Unit
tPHL			
Clock to output			
(CL = 150 pF)	18 min,	20 min,	nS
	610 max	650 max	nS
Set to output			
(CL = 50 pF)	12 min,	12 min,	nS
	385 max	412 max	nS
(CL = 150 pF)	18 min,	20 min,	nS
	610 max	650 max	nS
Reset to output			
(CL = 50 pF)	15 min,	16 min,	nS
	500 max	531 max	nS
(CL = 150 pF)	22 min,	23 min,	nS
	720 max	767 max	nS
VDD = 10.0V (CL = 50 pF)			
tPLH			
Clock to output	5 min,	5 min,	nS
	165 max	177 max	nS
Set to Q or Reset to $\overline{Q}$	5 min,	5 min,	nS
	165 max	177 max	nS
tPHL			
Clock to output	5 min,	5 min,	nS
	165 max	177 max	nS
Set to Q or Reset to $\overline{Q}$	5 min,	5 min,	nS
	165 max	177 max	nS
VDD = 15.0V (CL = 50 pF)			
tPLH			
Clock to output	3 min,	4 min,	nS
	110 max	120 max	nS
Set to Q or Reset to $\overline{Q}$	3 min,	4 min,	nS
	110 max	120 max	nS
tPHL			
Clock to output	3 min,	4 min,	nS
	110 max	120 max	nS
Set to Q or Reset to $\overline{Q}$	3 min,	4 min,	nS
	110 max	120 max	nS

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	Class I	Class II	Unit
Clock input			
Frequency (VDD = 5.0V, toggle mode)	1.5 min	1.5 min	MHz
Pulse width	330 max	330 max	nS
Set or Reset pulse width	250 max	250 max	nS
Input times			
Setup time (VDD = 5.0V)	200 max	200 max	nS
Hold time (VDD = 5.0V)	0 max	0 max	nS

6.4 Module function. This module contains five independent CMOS JK flip-flops and two 220 pF capacitors with one side connected to VSS, for external use.

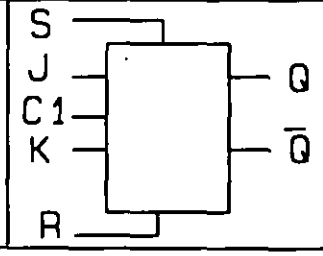
6.4.1 Input to output relationship. Each flip-flop has the input to output relationship in table II for positive logic.

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TABLE II. Input to output relationship for each JK-flip-flop.

INPUTS						OUTPUTS *	
C1 +	J	K	S	R	QN ‡	QN+1	$\overline{QN+1}$
⌐	H	X	L	L	L	H	L
⌐	X	L	L	L	H	H	L
⌐	L	X	L	L	L	L	H
⌐	X	H	L	L	H	L	H
⌐	X	X	L	L	X	Qn	$\overline{Qn}$
X	X	X	H	L	X	H	L
X	X	X	L	H	X	L	H
X	X	X	H	H	X	H	H

X=DON'T CARE
+ =LEVEL CHANGE
‡ =PRESENT STATE
*=NEXT STATE
H=HIGH LEVEL
L=LOW LEVEL



6.4.2 Paralleling outputs. Paralleling of outputs to obtain extended drive is permissible only when parallel inputs and outputs are confined within each group specified below. The output drive capability of this combination is equal to the sum of the output drive capabilities of the individual circuits. The circuit groups which can be paralleled are:

(a) 2 and 3
(b) 4 and 5

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6.4.3 Module handling considerations. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedures should be followed:

(a) Device pins should be in contact with conductive material, except when being tested or in actual operation, to avoid build-up of static charge.
(b) Soldering iron tips, metal parts of fixtures and tools, and handling facilities should be grounded.
(c) Signals should not be applied to the inputs while the device power supply is off.
(d) Input voltages greater than the supply (VDD - VSS) should not be applied as this will forward bias the protection diodes.
(e) All unused input leads must be connected to either VSS (ground) or VDD (device supply), whichever is appropriate for the logic circuit involved.

6.4.4 Dynamic power dissipation.

The dynamic power dissipation can be approximated by the formula:

$$PTOTAL = (C\ VDD^2\ f)\ N + \text{quiescent power dissipation}$$

where

C = load capacitance.
VDD = supply voltage.
f = operating frequency.
N = number of outputs.

6.5 CAUTION. This caution applies to each test figure. Failure to comply with the following procedures will result in irreversible module damage:

All test equipment must be grounded.
VDD must be applied before applying input signals or condition voltages.
All input signals and conditioning voltages must be removed before VDD is disconnected.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders

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and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - ER
Navy - NW
Air Force - 85

Review activites:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

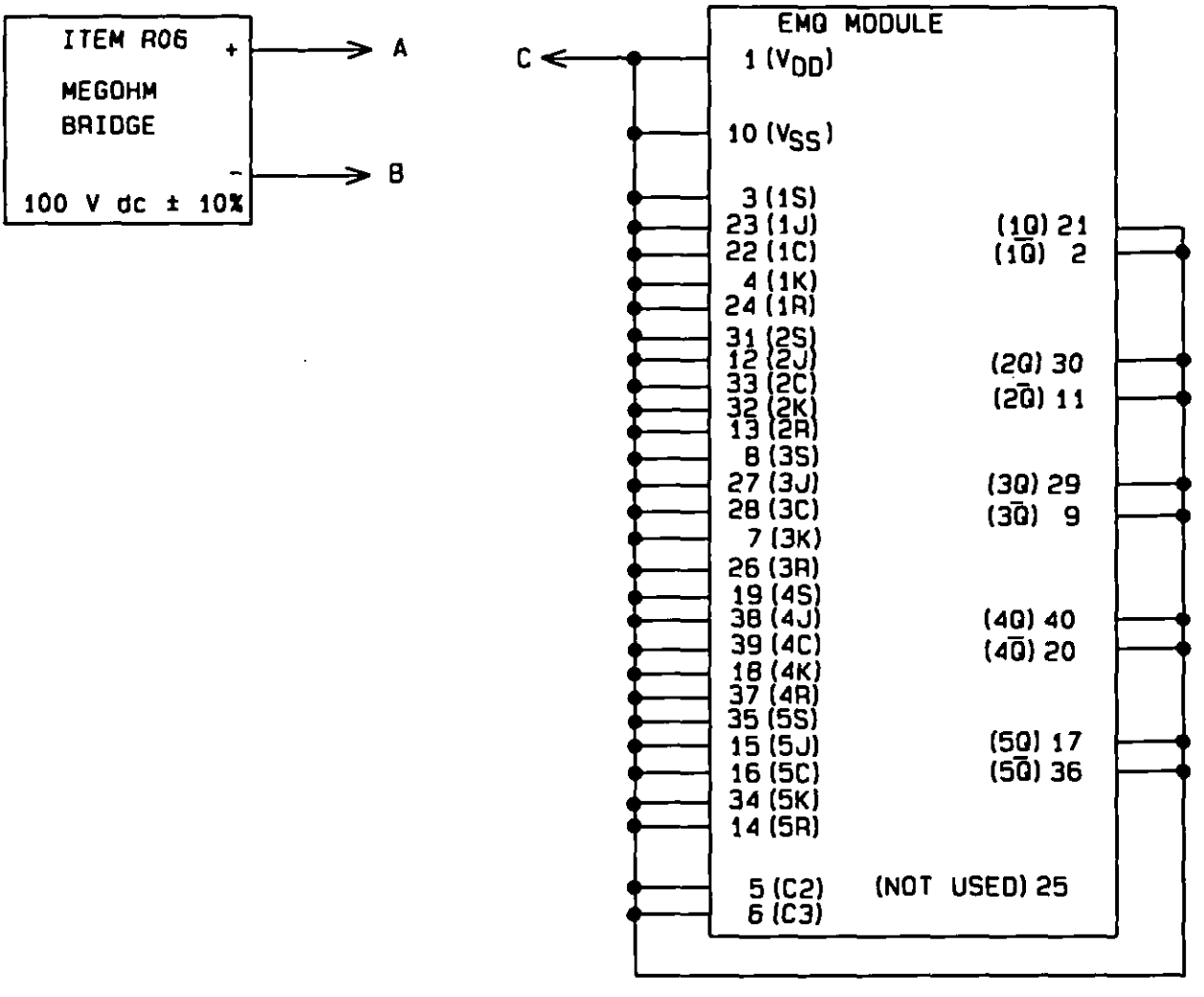
(Project 5963-0215)

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TABLE III. Isolation test.

Test cond	Parameter	Connect		Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
A1	Isolation	C, pin 25	Frame	10	INF	MOhm	5
A2	Isolation	pin 25	C	10	INF	MOhm	5

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition A1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5. Isolation test circuit for table III.

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TABLE IV. Negative input clamp diode voltage test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
B1	1S Negative input clamp diode voltage	3	Open	-0.4	-6.0	V	6
B2	1J	23					
B3	1C	22					
B4	1K	4					
B5	1R	24					
B6	2S	31					
B7	2J	12					
B8	2C	33					
B9	2K	32					
B10	2R	13					
B11	3S	8					
B12	3J	27					
B13	3C	28					
B14	3K	7					
B15	3R	26					
B16	4S	19					
B17	4J	38					
B18	4C	39					
B19	4K	18					
B20	4R Negative input clamp diode voltage	37	Open	-0.4	-6.0	V	6

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TABLE IV. Negative input clamp diode voltage test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
B21	5S Negative input clamp diode voltage	35	Open	-0.4	-6.0	V	6
B22	5J	15					
B23	5C	16					
B24	5K	34					
B25	5R Negative input clamp diode voltage	14	Open	-0.4	-6.0	V	6

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TABLE V. Positive input clamp diode voltage test.

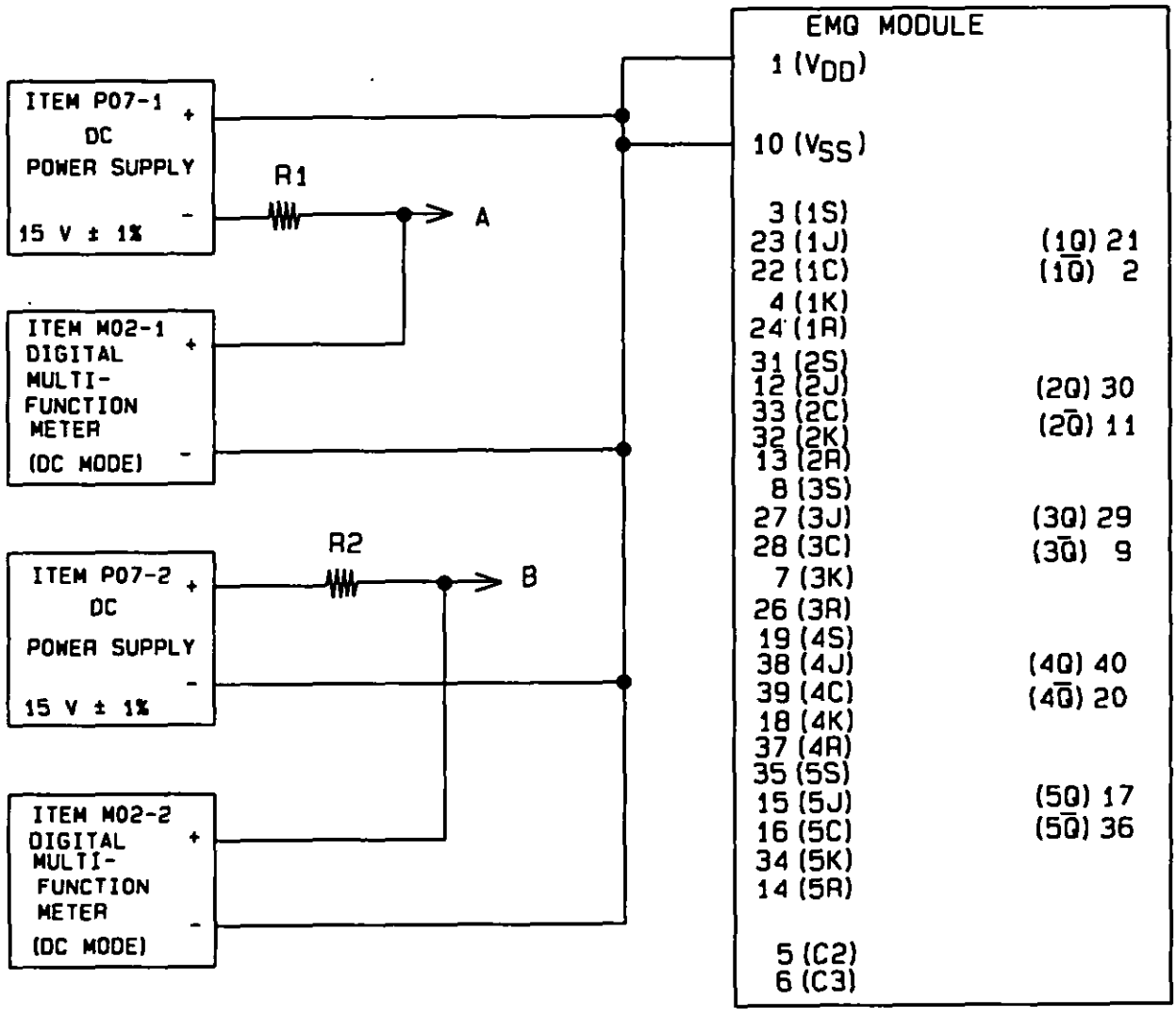
Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
C1	1S Positive input clamp diode voltage	Open	3	0.4	1.5	V	6
C2	1J		23				
C3	1C		22				
C4	1K		4				
C5	1R		24				
C6	2S		31				
C7	2J		12				
C8	2C		33				
C9	2K		32				
C10	2R		13				
C11	3S		8				
C12	3J		27				
C13	3C		28				
C14	3K		7				
C15	3R		26				
C16	4S		19				
C17	4J		38				
C18	4C		39				
C19	4K		18				
C20	4R Positive input clamp diode voltage	Open	37	0.4	1.5	V	6

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TABLE V. Positive input clamp diode voltage test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
C21	5S Positive input clamp diode voltage	Open	35	0.4	1.5	V	6
C22	5J		15				
C23	5C		16				
C24	5K		34				
C25	5R Positive input clamp diode voltage	Open	14	0.4	1.5	V	6

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NOTES:

1. R1 = 9.09 kOhms ± 1 percent, 1/8 W.
2. R2 = 13.7 kOhms ± 1 percent, 1/8 W.
3. Since items P07-1, P07-2, M02-1, and M02-2 are not used simultaneously, only one item P07 and item M02 is required.

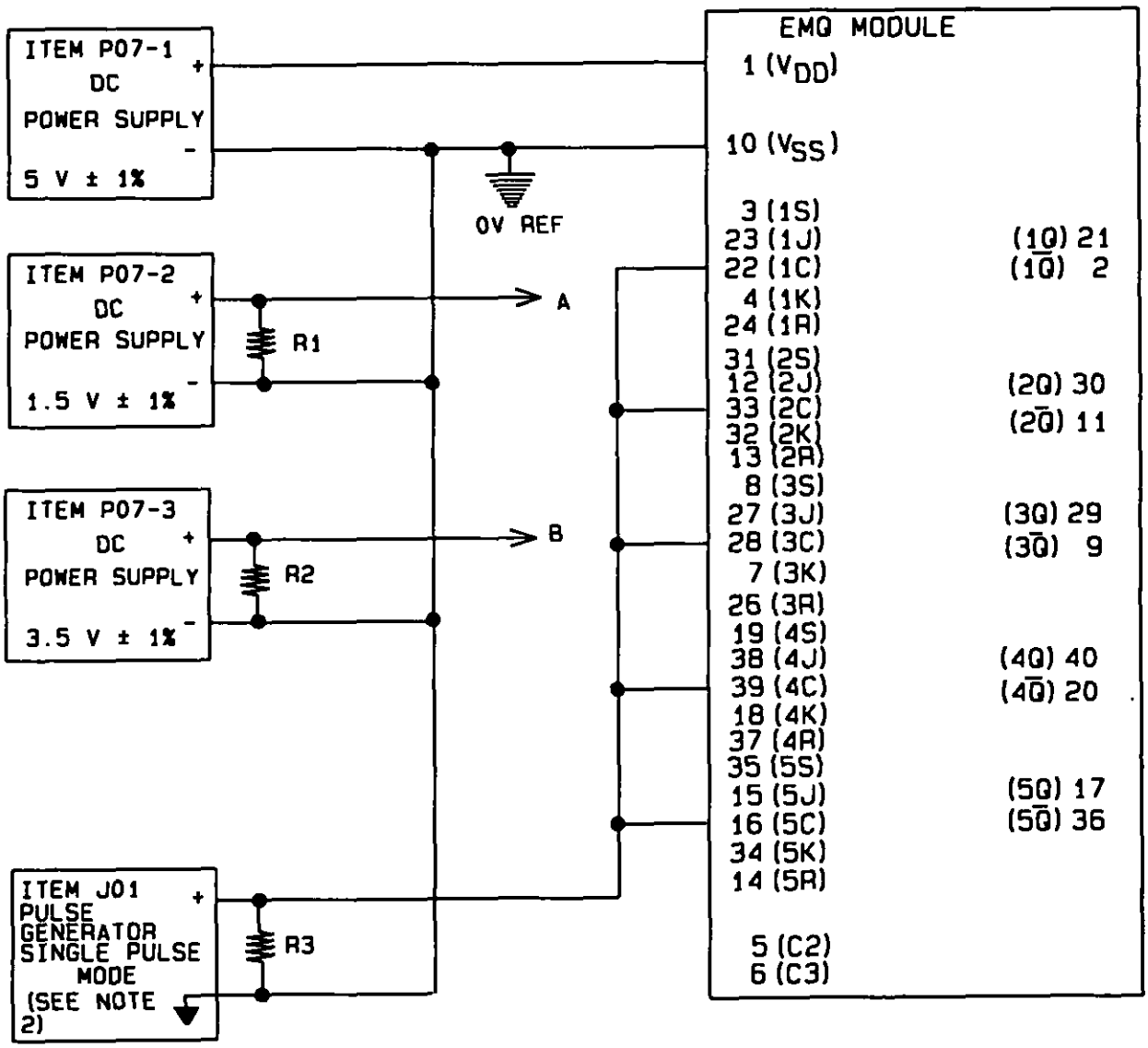
FIGURE 6. Negative and positive input clamp diode voltage test for tables IV and V.

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TABLE VI. Functional test.

Test cond	Parameter	Connect		Output logic state										Test figure
		A to pin	B to pin	Q1	Q2	Q3	Q4	Q5	$\overline{Q1}$	$\overline{Q2}$	$\overline{Q3}$	$\overline{Q4}$	$\overline{Q5}$	
D1	Functional	4,24,32,13,7,26,18, 37,34,14	3,23,31,12,8,27,19,38, 35,15	H	H	H	H	H	L	L	L	L	L	7
D2		3,4,31,32,8,7,19,18, 35,34	23,24,12,13,27,26,38, 37,15,14	L	L	L	L	L	H	H	H	H	H	
D3		3,4,24,31,32,13,8,7, 26,19,18,37,35,34,14	23,12,27,38,15	H	H	H	H	H	L	L	L	L	L	
D4		3,23,24,31,12,13,8, 27,26,19,38,37,35,15, 14	4,32,7,18,34	L	L	L	L	L	H	H	H	H	H	
D5		3,24,31,13,8,26,19, 37,35,14	23,4,12,32,27,7,38,18, 15,34	H	H	H	H	H	L	L	L	L	L	
D6		3,24,31,13,8,26,19, 37,35,14	23,4,12,32,27,38,18,15, 34	L	L	L	L	L	H	H	H	H	H	
D7	Functional	3,23,4,24,31,12,32, 13,8,27,7,28,19,38, 18,37,35,15,34,14	Open	L	L	L	L	L	H	H	H	H	H	7

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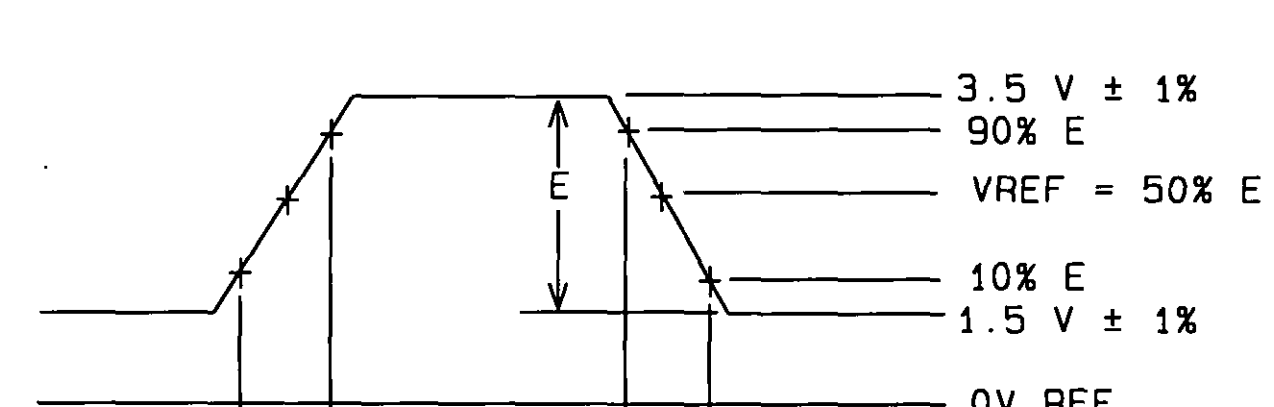


NOTES:

1. R1 = R2 = R3 = 51.1 Ohms ± 5 percent, 1/2 W.
2. After the input conditions are setup for each test condition, apply one pulse using item J01, then make the required measurement. (Test conditions D1-D7 must be done in sequence without any power interruption to VDD).

FIGURE 7. Functional test circuit for table VI.

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3. Item J01 setup:
PW = 1 μS ± 10 percent.
tTLH = tTHL = 10 ± 2 nS.
PRR = 500 kHz ± 10 percent. (The PRR is used for setup procedures only. When the pulse has been setup, item J01 should be placed in the single cycle mode.)

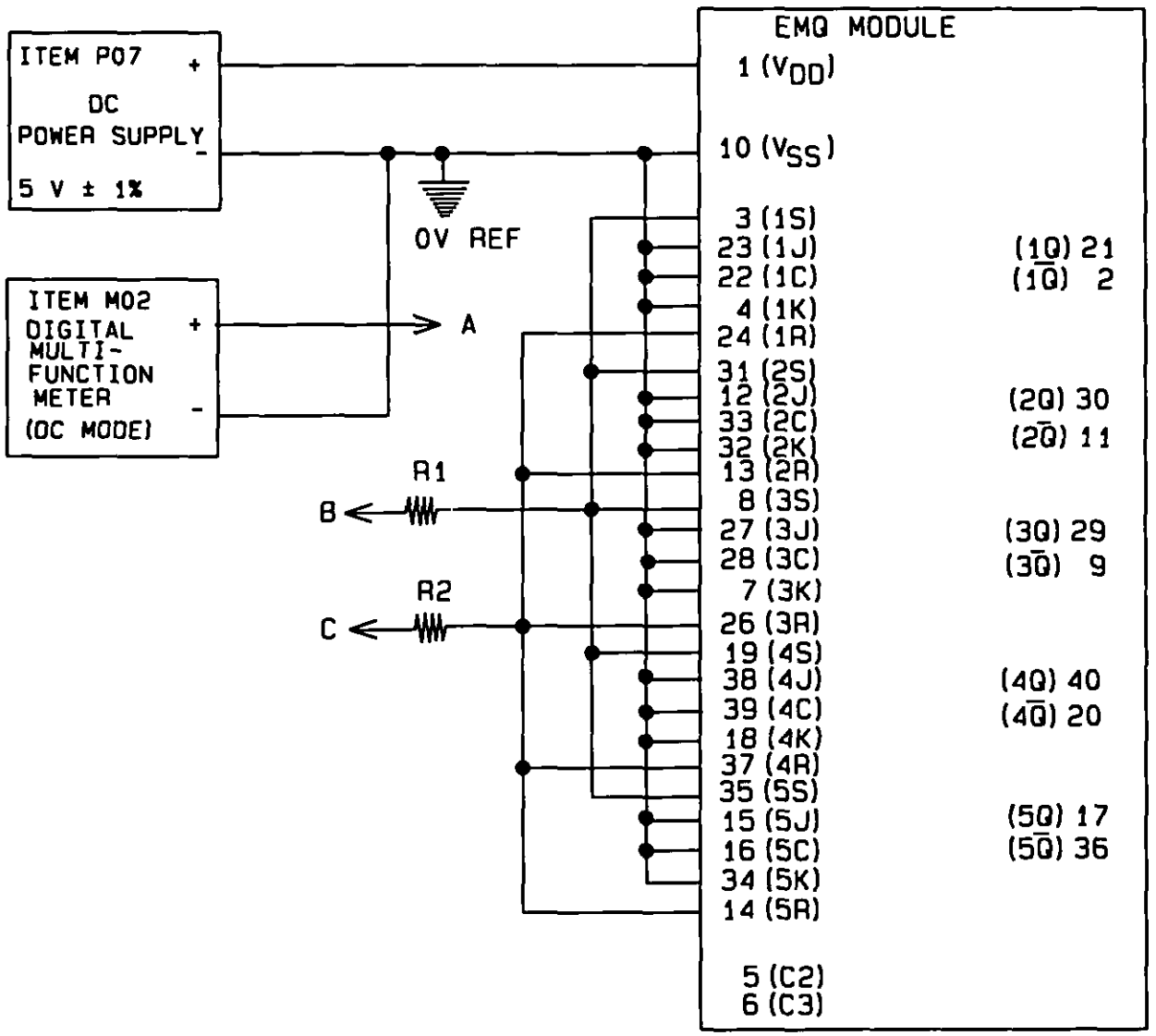
FIGURE 7. Functional test circuit for table VI - Continued.

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TABLE VII. Output voltage (low-level and high-level) test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test figure
E1	1 Output voltage, (IOL and IOH = 0) Low level	21	10	1	0	50	mV	8
F1	1 High level	2			4.95	5.05	V	
E2	2 Low level	30			0	50	mV	
F2	2 High level	11			4.95	5.05	V	
E3	3 Low level	29			0	50	mV	
F3	3 High level	9			4.95	5.05	V	
E4	4 Low level	40			0	50	mV	
F4	4 High level	20			4.95	5.05	V	
E5	5 Low level	17			0	50	mV	
F5	5 High level	36	10	1	4.95	5.05	V	
F6	6 High level	21	1	10	4.95	5.05	V	
E6	8 Low level	2			0	50	mV	
F7	7 High level	30			4.95	5.05	V	
E7	7 Low level	11			0	50	mV	
F8	8 High level	29			4.95	5.05	V	
E8	8 Low level	9			0	50	mV	
F9	9 High level	40			4.95	5.05	V	
E9	9 Low level	20			0	50	mV	
F10	10 High level	17			4.95	5.05	V	
E10	10 Output voltage (IOL and IOH = 0) Low level	36	1	10	0	50	mV	8

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NOTE:
R1 = R2 = 1.5 kOhms ± 1 percent, 1/8 W.

FIGURE 8. Output voltage (low-level and high-level) test circuit for table VII.

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TABLE VIII. Output voltage (low-level) test.

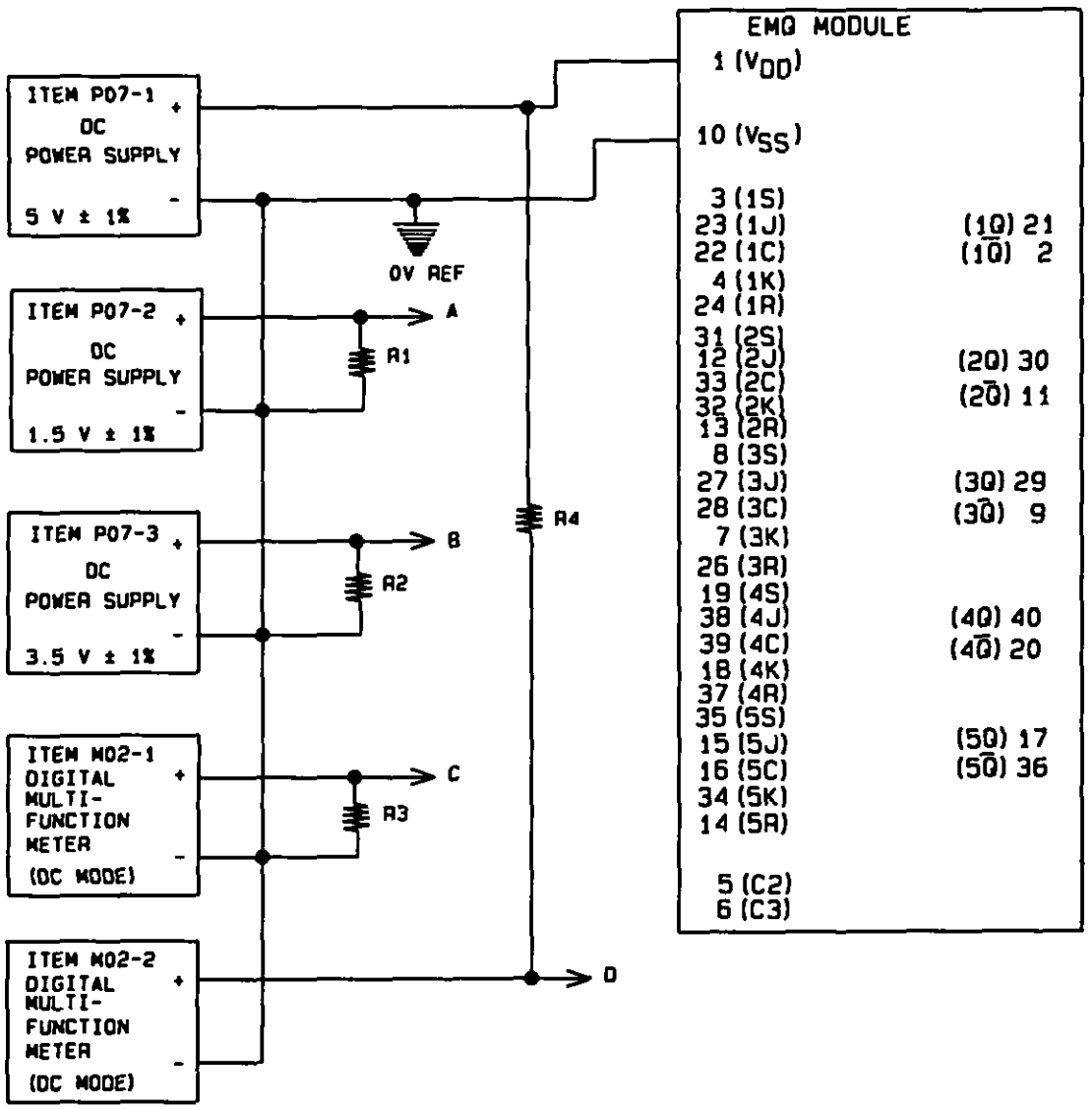
Test cond	Parameter	Connect				Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	1Q Output voltage, low level, (VDD = 5 V)	3,23,22,4,12,33,32,13, 27,28,7,26,38,39,18, 37,15,16,34,14	24,31,8,19,35	Open	21	0	400	mV	9
G2	2Q	23,22,4,24,31,12,33, 32,27,28,7,26,38,39, 18,37,15,16,34,14	3,13,8,19,35		30				
G3	3Q	23,22,4,24,12,33,32, 13,8,27,28,7,38,39, 18,37,15,16,34,14	3,31,26,19,35		29				
G4	4Q	23,22,4,24,12,33,32, 13,27,28,7,26,19,38, 39,18,15,16,34,14	3,31,8,37,35		40				
G5	5Q	23,22,4,24,12,33,32, 13,27,28,7,26,38,39, 18,37,35,15,16,34	3,31,8,19,14		17				
G6	1$\bar{Q}$	24,31,8,19,35	3,23,22,4,12,33,32, 13,27,28,7,26,38,39, 18,37,15,16,34,14		2				
G7	2$\bar{Q}$	3,13,8,19,35	23,22,4,24,31,12,33, 32,27,28,7,26,38,39, 18,37,15,16,34,14		11				
G8	3$\bar{Q}$	3,31,26,19,35	23,22,4,24,12,33,32, 13,8,27,28,7,38,39, 18,37,15,16,34,14		9				
G9	4$\bar{Q}$	3,31,8,37,35	23,22,4,24,12,33,32, 13,27,28,7,26,19,38, 39,18,15,16,34,14		20				
G10	5$\bar{Q}$ Output voltage, low level (VDD = 5 V)	3,31,8,19,14	23,22,4,24,12,33,32, 13,27,28,7,26,38,39, 18,37,35,15,16,34	Open	36	0	400	mV	9

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TABLE IX. Output voltage (high-level) test.

Test cond	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25° ± 5°C ambient: Min	Initial limits 25° ± 5°C ambient: Max	Unit	Test fig
H1	1Q Output voltage, high level, (VDD = 5 V)	24,31,8,19,35	3,23,22,4,12,33,32, 13,27,28,7,26,38,39, 18,37,15,16,34,14	21	Open	4.6	5.0	V	9
H2	2Q	3,13,8,19,35	23,22,4,24,31,12,33, 32,27,28,7,26,38,39, 18,37,15,16,34,14	30					
H3	3Q	3,31,26,19,35	23,22,4,24,12,33,32, 13,8,27,28,7,38,39, 18,37,15,16,34,14	29					
H4	4Q	3,31,8,37,35	23,22,4,24,12,33,32, 13,27,28,7,26,19,38, 39,18,15,16,34,14	40					
H5	5Q	3,31,8,19,14	23,22,4,24,12,33,32, 13,27,28,7,26,38,39, 18,37,35,15,16,34	17					
H6	1$\bar{Q}$	3,23,22,4,12,33,32,13, 27,28,7,26,38,39,18, 37,15,16,34,14	24,31,8,19,35	2					
H7	2$\bar{Q}$	23,22,4,24,31,12,33, 32,27,14,28,7,26,38, 39,18,37,15,16,34	3,13,8,19,35	11					
H8	3$\bar{Q}$	23,22,4,24,12,33,32, 13,8,27,28,7,38,39,18, 37,15,16,34,14	3,31,26,19,35	9					
H9	4$\bar{Q}$	23,22,4,24,12,33,32, 13,27,28,7,26,19,38, 39,18,15,16,34,14	3,31,8,37,35	20					
H10	5$\bar{Q}$ Output voltage, high level, (VDD = 5 V)	23,22,4,24,12,33,32, 13,27,28,7,26,38,39, 18,37,35,15,16,34	3,31,8,19,14	36	Open	4.6	5.0	V	9

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NOTES:

1. R1 = R2 = 51.1 Ohms ± 5 percent, 1/2 W.
2. R3 = At -55°C 7.15 kOhms ± 1 percent, 1/8 W.
   = At 0° and 25°C 9.09 kOhms ± 1 percent, 1/8 W.
   = At 60°C 10.5 kOhms ± 1 percent, 1/8 W.
   = At 85°C 12.7 kOhms ± 1 percent, 1/8 W.
3. R4 = At -55°C 7.15 kOhms ± 1 percent, 1/8 W.
   = At 0° and 25°C 9.09 kOhms ± 1 percent, 1/8 W.
   = At 60°C 10.5 kOhms ± 1 percent, 1/8 W.
   = At 85°C 12.7 kOhms ± 1 percent, 1/8 W.

* FIGURE 9. Output voltage (high-level and low-level) test circuit for tables VIII and IX.

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TABLE X. Output voltage (low-level) test.

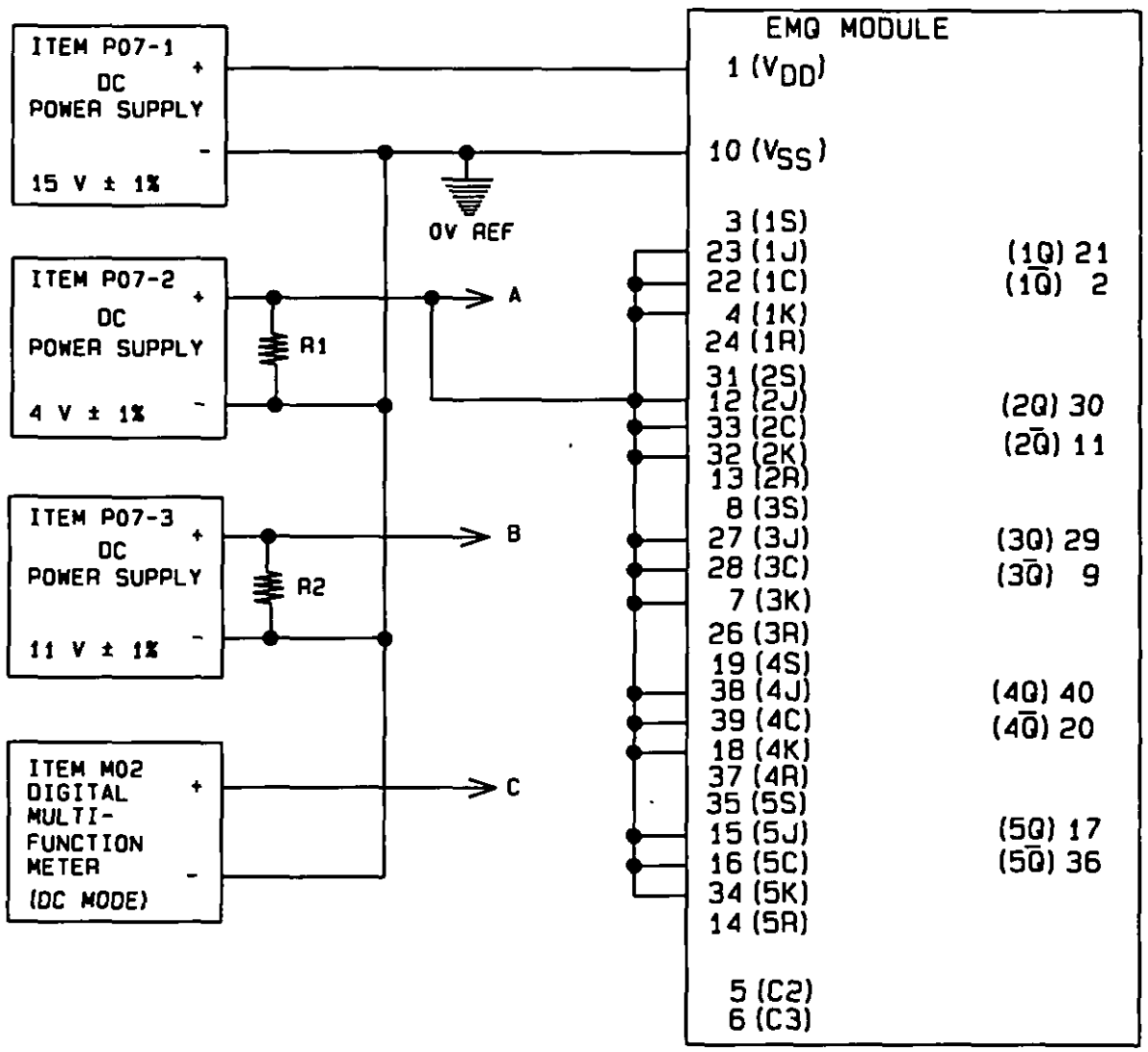
Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
J1	1Q Output voltage, low level, (VDD = 15 V)	3,31,8,19,35	24,13, 26,37, 14	21	0.0	1.5	V	10
J2	2Q			30				
J3	3Q			29				
J4	4Q			40				
J5	5Q	3,31,8,19,35	24,13, 26,37, 14	17				
J6	$1\bar{Q}$	24,13,26,37,14	3,31,8, 19,35	2				
J7	$2\bar{Q}$			11				
J8	$3\bar{Q}$			9				
J9	$4\bar{Q}$			20				
J10	$5\bar{Q}$ Output voltage, low level, (VDD = 15 V)	24,13,26,37,14	3,31,8, 19,35	36	0.0	1.5	V	10

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TABLE XI. Output voltage (high-level) test.

Test cond	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
K1	1Q	Output voltage, high level, (VDD = 15 V)	24,13,26,37,14	3,31,8, 19,35	21	13.5	15.0	V	10
K2	2Q				30				
K3	3Q				29				
K4	4Q				40				
K5	5Q		24,13,26,37,14	3,31,8, 19,35	17				
K6	1$\bar{Q}$		3,31,8,19,35	24,13, 26,37, 14	2				
K7	2$\bar{Q}$				11				
K8	3$\bar{Q}$				9				
K9	4$\bar{Q}$				20				
K10	5$\bar{Q}$	Output voltage, high level, (VDD = 15 V)	3,31,8,19,35	24,13, 26,37, 14	36	13.5	15.0	V	10

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:13Z
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NOTES:
1. R1 = 51.1 Ohms ± 5 percent, 1/2 W.
2. R2 = 576 Ohms ± 5 percent, 1/4 W.

FIGURE 10. Output voltage (low-level and high-level) test circuit for tables X and XI.

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TABLE XII. Input current (low-level) test.

Test cond	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25° ± 5°C ambient: Min	Initial limits 25° ± 5°C ambient: Max	Unit	Test fig
L1	1S Input current, low level	23,22,4,24,31,12,33,32,13,8, 27,28,7,26,19,38,39,18,37, 35,15,16,34,14	3	0	-200	nA	11
L2	1J	3,22,4,24,31,12,33,32,13,8, 27,28,7,26,19,38,39,18,37, 35,15,16,34,14	23				
L3	1C	3,23,4,24,31,12,33,32,13,8, 27,28,26,19,7,38,39,18,37, 35,15,16,34,14	22				
L4	1K	3,23,22,24,31,12,33,32,13,8, 27,28,7,26,19,38,39,18,37, 35,15,16,34,14	4				
L5	1R	3,23,22,4,31,12,33,32,13,8, 27,28,7,26,19,38,39,18,37, 35,15,16,34,14	24				
L6	2S	3,23,22,4,24,12,33,32,13,8, 27,28,26,19,7,38,39,18,37, 35,15,16,34,14	31				
L7	2J Input current, low level	3,23,22,4,24,31,33,32,13,8, 27,28,7,26,19,38,39,18,37, 35,15,16,34,14	12	0	-200	nA	11

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TABLE XII. Input current (low-level) test - Continued.

Test cond	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25° ± 5°C ambient: Min	Initial limits 25° ± 5°C ambient: Max	Unit	Test fig
L8	2C Input current, low level	3,23,22,4,24,31,12,32,13,8, 27,28,7,26,19,38,39,18,37, 35,15,16,34,14	33	0	-200	nA	11
L9	2K	3,23,22,4,24,31,12,33,13,8, 27,28,7,26,19,38,39,18,37, 35,15,16,34,14	32				
L10	2R	3,23,22,4,24,31,12,33,32,8, 27,28,7,26,19,38,39,18,37, 35,15,16,34,14	13				
L11	3S	3,23,22,4,24,31,12,33,32,13, 27,28,7,26,19,38,39,18,37, 35,15,16,34,14	8				
L12	3J	3,23,22,4,24,31,12,33,32,13, 8,28,7,26,19,38,39,18,37,35, 15,16,34,14	27				
L13	3C	3,23,22,4,24,31,12,33,32,13, 8,27,7,26,19,38,39,18,37,35, 15,16,34,14	28				
L14	3K	3,23,22,4,24,31,12,33,32,13, 8,27,28,26,19,38,39,18,37, 35,15,16,34,14	7				
L15	3R Input current, low level	3,23,22,4,24,31,12,33,32,13, 8,27,28,7,19,38,39,18,37,35, 15,16,34,14	26	0	-200	nA	11

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:13Z
Check the source to verify that this is the current version before use.

TABLE XII. Input current (low-level) test - Continued.

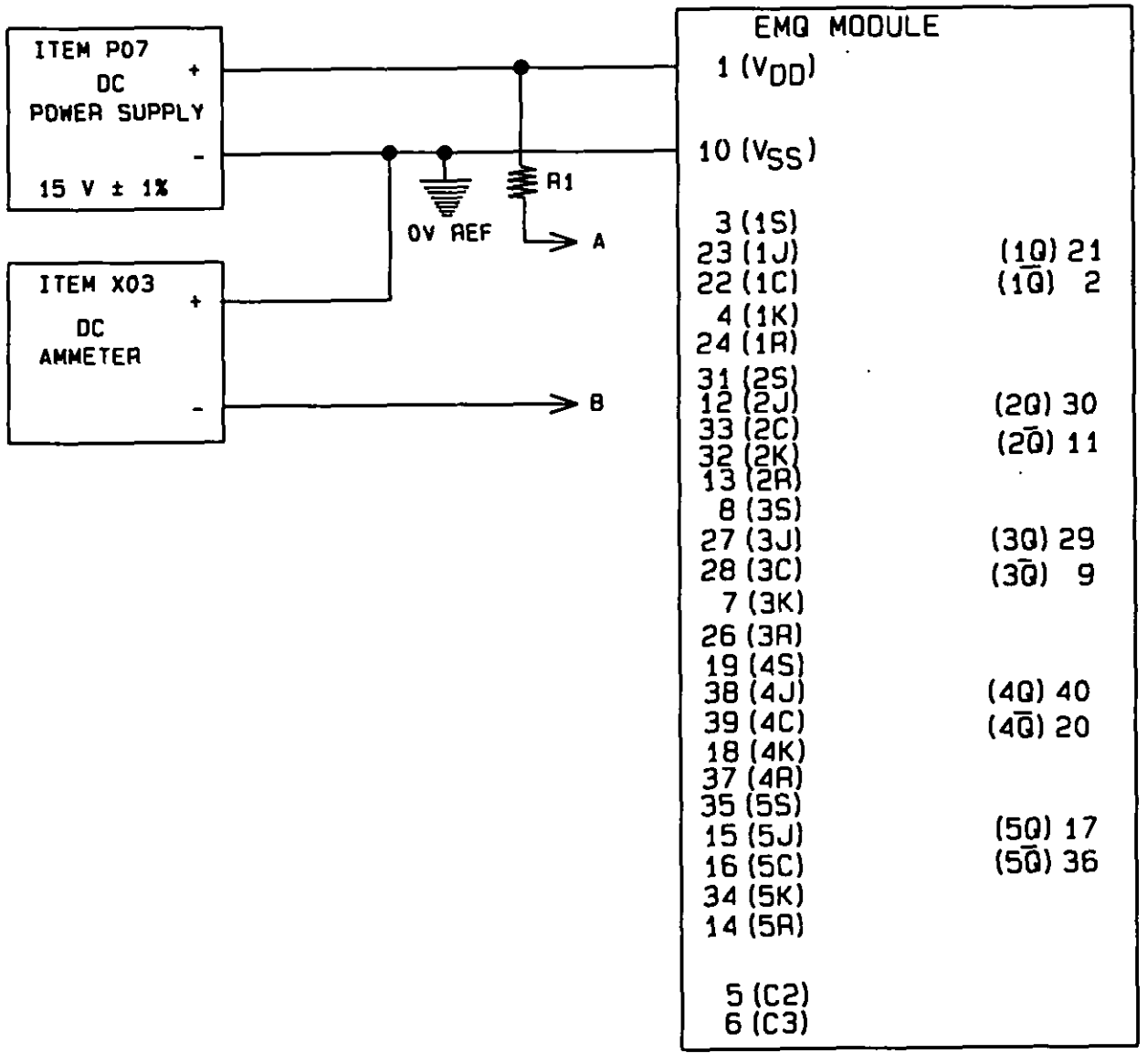
Test cond	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25° ± 5°C ambient: Min	Initial limits 25° ± 5°C ambient: Max	Unit	Test fig
L16	4S Input current, low level	3,23,22,4,24,31,12,33,32,13, 8,27,28,7,26,38,39,18,37,35, 15,16,34,14	19	0	-200	nA	11
L17	4J	3,23,22,4,24,31,12,33,32,13, 8,27,28,7,26,19,39,18,37,35, 15,16,34,14	38				
L18	4C	3,23,22,4,24,31,12,33,32,13, 8,27,28,7,26,19,38,18,37,35, 15,16,34,14	39				
L19	4K	3,23,22,4,24,31,12,33,32,13, 8,27,28,7,26,19,38,39,37,35, 15,16,34,14	18				
L20	4R	3,23,22,4,24,31,12,33,32,13, 8,27,28,7,26,19,38,39,18,35, 15,16,34,14	37				
L21	5S	3,23,22,4,24,31,12,33,32,13, 8,27,28,7,26,19,38,39,18,37, 15,16,34,14	35				
L22	5J	3,23,22,4,24,31,12,33,32,13, 8,27,28,7,26,19,38,39,18,37, 35,16,34,14	15				
L23	5C Input current, low level	3,23,22,4,24,31,12,33,32,13, 8,27,28,7,26,19,38,39,18,37, 35,15,34,14	16	0	-200	nA	11

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:13Z
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TABLE XII. Input current (low-level) test - Continued.

Test cond	Parameter	Connect		Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
L24	5K Input current, low level	3,23,22,4,24,31,12,33,32,13,8,27,28,7,26,19,38,39,18,37,35,15,16,14	34	0	-200	nA	11
L25	5R Input current, low level	3,23,22,4,24,31,12,33,32,13,8,27,28,7,26,19,38,39,18,37,35,15,16,34	14	0	-200	nA	11

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CAUTION:
Do not isolate the chassis of item X03 from earth ground.

NOTE: R1 = 1.5 kOhms ± 1 percent, 1/4 W.

* FIGURE 11. Input current (low-level) test circuit for table XII.

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TABLE XIII. Input current (high-level) test.

Test cond	Parameter		Connect: A to pin	Connect: B to pin	Initial limits 25° ± 5°C ambient: Min	Initial limits 25° ± 5°C ambient: Max	Unit	Test fig
M1	1S	Input current, high level	3	23,22,4,24,31,12,33,32,13, 8,27,28,7,26,19,38,39,18, 37,35,15,16,34,14	0	200	nA	12
M2	1J		23	3,22,4,24,31,12,33,32,13, 8,27,28,7,26,19,38,39,18, 37,35,15,16,34,14				
M3	1C		22	3,23,4,24,31,12,33,32,13, 8,27,28,7,26,19,38,39,18, 37,35,16,34,14,15				
M4	1K		4	3,23,22,24,31,12,33,32,13, 8,27,28,7,26,19,38,39,18, 37,35,15,16,34,14				
M5	1R		24	3,23,22,4,31,12,33,32,13, 8,27,28,7,26,19,38,39,18, 37,35,15,16,34,14				
M6	2S		31	3,23,22,4,24,12,33,32,13, 8,27,28,7,26,19,38,39,18, 37,35,15,16,34,14				
M7	2J		12	3,23,22,4,24,31,33,32,13, 8,27,28,7,26,19,38,39,18, 37,35,15,16,34,14				
M8	2C	Input current, high level	33	3,23,22,4,24,31,12,32,13, 8,27,28,7,26,19,38,39,18, 37,35,15,16,34,14	0	200	nA	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:13Z
Check the source to verify that this is the current version before use.

TABLE XIII. Input current (high-level) test - Continued.

Test cond	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25° ± 5°C ambient: Min	Initial limits 25° ± 5°C ambient: Max	Unit	Test fig
M9	2K Input current, high level	32	3,23,22,4,24,31,12,33,13, 8,27,28,7,26,19,38,39,18, 37,35,15,16,34,14	0	200	nA	12
M10	2R	13	3,23,22,4,24,31,12,33,32, 8,27,28,7,26,19,38,39,18, 37,35,15,16,34,14				
M11	3S	8	3,23,22,4,24,31,12,33,32, 13,28,27,7,26,19,38,39,18, 37,35,15,16,34,14				
M12	3J	27	3,23,22,4,24,31,12,33,32, 13,8,28,7,26,19,38,39,18, 37,35,15,16,34,14				
M13	3C	28	3,23,22,4,24,31,12,33,32, 13,8,27,7,26,19,38,39,18, 37,35,15,16,34,14				
M14	3K	7	3,23,22,4,24,31,12,33,32, 13,8,27,28,26,19,38,39,18, 37,35,15,16,34,14				
M15	3R	26	3,23,22,4,24,31,12,33,32, 13,8,27,28,7,19,38,39,18, 37,35,15,16,34,14				
M16	4S Input current, high level	19	3,23,22,4,24,31,12,33,32, 13,8,27,28,7,26,38,39,18, 37,35,15,16,34,14	0	200	nA	12

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:13Z
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TABLE XIII. Input current (high-level) test - Continued.

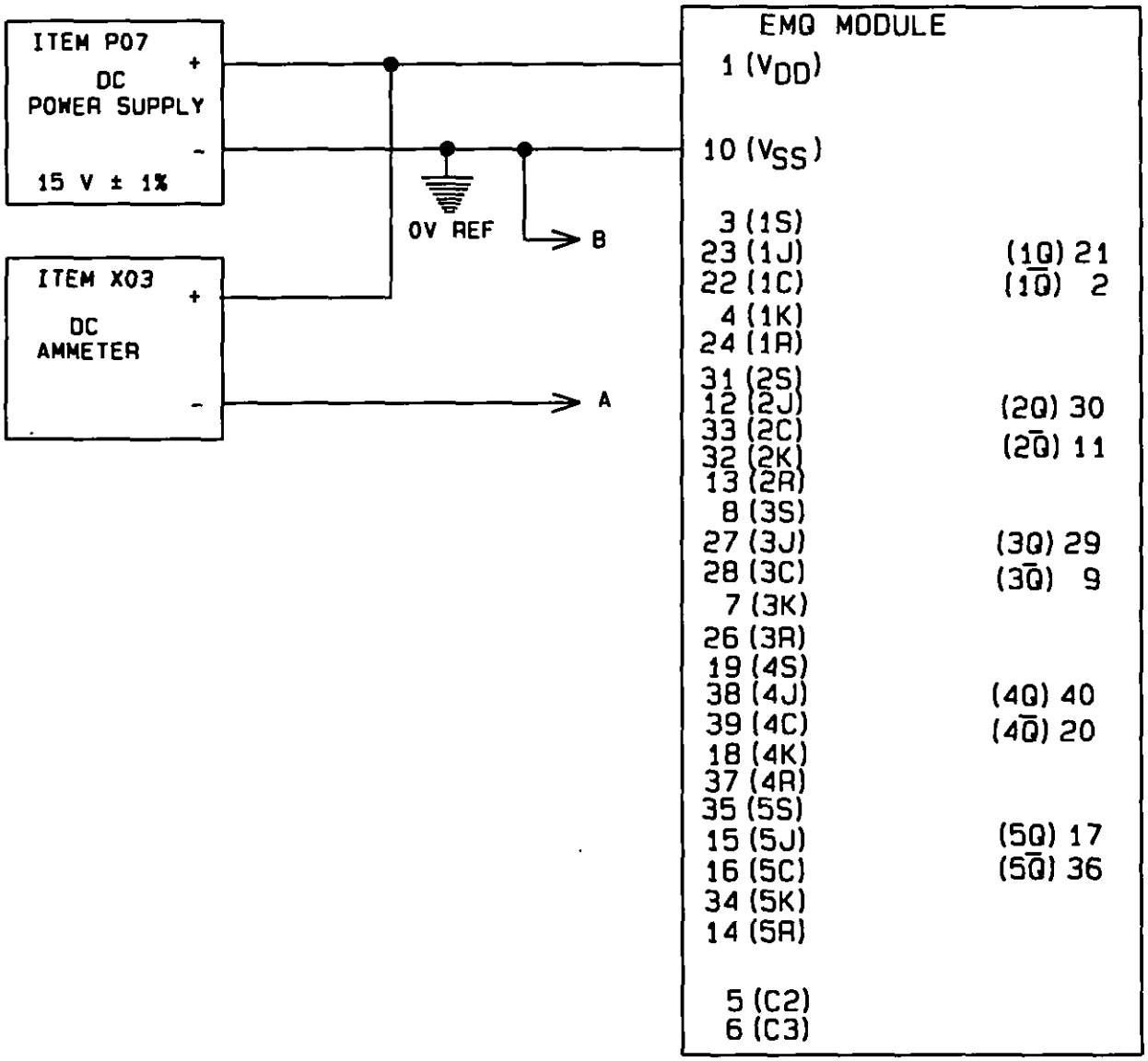
Test cond		Parameter	Connect A to pin	Connect B to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
M17	4J	Input current, high level	38	3,23,22,4,24,31,12,33,32, 13,8,27,28,7,26,19,39,18, 37,35,15,16,34,14	0	200	nA	12
M18	4C		39	3,23,22,4,24,31,12,33,32, 13,8,27,28,7,26,19,38,18, 37,35,15,16,34,14				
M19	4K		18	3,23,22,4,24,31,12,33,32, 13,8,27,28,7,26,19,38,39, 37,35,15,16,34,14				
M20	4R		37	3,23,22,4,24,31,12,33,32, 13,8,27,28,7,26,19,38,39, 18,35,15,16,34,14				
M21	5S		35	3,23,22,4,24,31,12,33,32, 13,8,27,28,7,26,19,38,39, 18,37,15,16,34,14				
M22	5J		15	3,23,22,4,24,31,12,33,32, 13,8,27,28,7,26,19,38,39, 18,37,35,16,34,14				
M23	5C		16	3,23,22,4,24,31,12,33,32, 13,8,27,28,7,26,19,38,39, 18,37,35,15,34,14				
M24	5K	Input current, high level	34	3,23,22,4,24,31,12,33,32, 13,8,27,28,7,26,19,38,39, 18,37,35,15,16,14	0	200	nA	12

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TABLE XIII. Input current (high-level) test - Continued.

Test cond	Parameter	Connect		Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
M25	5R Input current, high level	14	3,23,22,4,24,31,12,33,32, 13,8,27,28,7,26,19,38,39, 18,37,35,15,16,34	0	200	nA	12

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CAUTION:
Do not isolate the chassis of item X03 from earth ground.

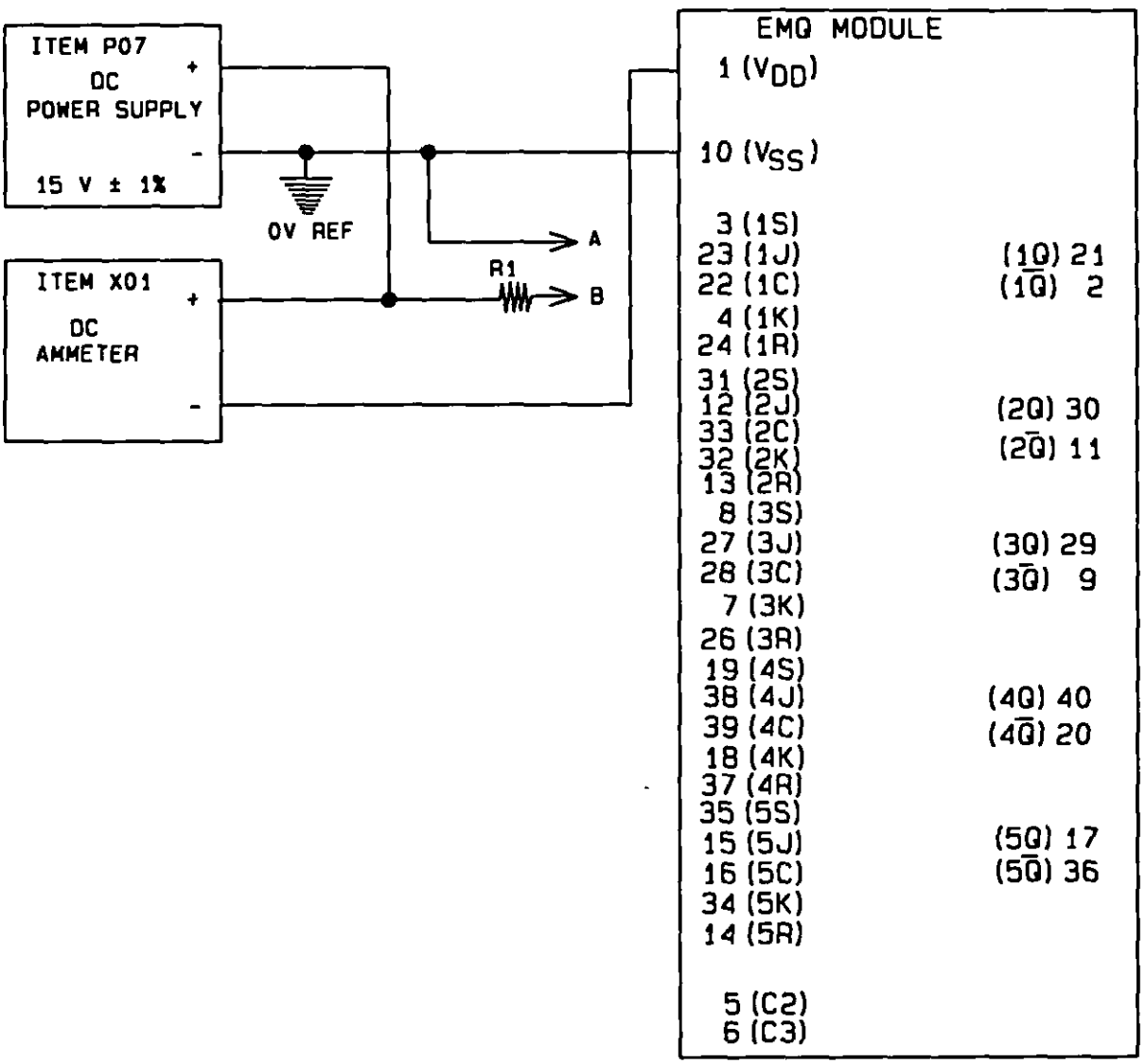
* FIGURE 12. Input current (high-level) test circuit for table XIII.

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TABLE XIV. Power supply current (low and high-level) test.

Test cond	Parameter	Connect		Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
N1	Power supply current, low level	3,31,8,19,35	23,22,4,24,12, 33,32,13,27,28, 7,26,38,39,18, 37,15,16,34,14	0	60	µA	13
N2	Power supply current, high level	23,22,4,24,12, 33,32,13,27,28, 7,26,38,39,18, 37,15,16,34,14	3,31,8,19,35	0	60	µA	13

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CAUTION:
Do not isolate the chassis of item X01 from earth ground.

NOTE: R1 = 1.5 kOhms ± 1 percent, 1/4 W.

FIGURE 13. Power supply current (low-level and high-level) test circuit for table XIV.

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• TABLE XV. Transition time (tTLH and tTHL) test.

Test cond	Parameter		Connect A to pin	B to pin	C to pin	D to pin	Initial limits 25° ± 5°C ambient Min	Max	Unit	Test fig
P1	1Q Transition time, CL = 50 pF, VDD = 5V	tTLH	12,33,27,28,38,39,15,16	23	22	21	11	360	nS	14,15
Q1	1$\bar{Q}$	tTHL	12,33,27,28,38,39,15,16	23	22	2	6	200		
P2	2Q	tTLH	23,22,27,28,38,39,15,16	12	33	30	11	360		
Q2	2$\bar{Q}$	tTHL	23,22,27,28,38,39,15,16	12	33	11	6	200		
P3	3Q	tTLH	23,22,12,33,38,39,15,16	27	28	29	11	360		
Q3	3$\bar{Q}$	tTHL	23,22,12,33,38,39,15,16	27	28	9	6	200		
P4	4Q	tTLH	23,22,12,33,27,28,15,16	38	39	40	11	360		
Q4	4$\bar{Q}$	tTHL	23,22,12,33,27,28,15,16	38	39	20	6	200		
P5	5Q	tTLH	23,22,12,33,27,28,38,39	15	16	17	11	360		
Q5	5$\bar{Q}$	tTHL	23,22,12,33,27,28,38,39	15	16	36	6	200		
Q6	1Q	tTHL	12,33,27,28,38,39,15,16	23	22	21	6	200		
P6	1$\bar{Q}$	tTLH	12,33,27,28,38,39,15,16	23	22	2	11	360		
Q7	2Q	tTHL	23,22,27,28,38,39,15,16	12	33	30	6	200		
P7	2$\bar{Q}$	tTLH	23,22,27,28,38,39,15,16	12	33	11	11	360		
Q8	3Q Transition time, CL = 50 pF, VDD = 5V	tTHL	23,22,12,33,38,39,15,16	27	28	29	6	200	nS	14,15

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*TABLE XV. Transition time (tTLH and tTHL) test - Continued.

Test cond		Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
P8	3$\bar{Q}$	Transition time, CL = 50 pF, VDD = 5V	tTLH	23,22,12,33,38,39,15,16	27	28	9	11	360	nS	14,15
Q9	4Q		tTHL	23,22,12,33,27,28,15,16	38	39	40	6	200		
P9	4$\bar{Q}$		tTLH	23,22,12,33,27,28,15,16	38	39	20	11	360		
Q10	5Q		tTHL	23,22,12,33,27,28,38,39	15	16	17	6	200		
P10	5$\bar{Q}$	Transition time, CL = 50 pF, VDD = 5V	tTLH	23,22,12,33,27,28,38,39	15	16	36	11	360		
P11	1Q	Transition time CL = 150 pF, VDD = 5V	tTLH	12,33,27,28,38,39,15,16	23	22	21	25	840		
Q11	1$\bar{Q}$		tTHL	12,33,27,28,38,39,15,16	23	22	2	20	680		
P12	2Q		tTLH	23,22,27,28,38,39,15,16	12	33	30	25	840		
Q12	2$\bar{Q}$		tTHL	23,22,27,28,38,39,15,16	12	33	11	20	680		
P13	3Q		tTLH	23,22,12,33,38,39,15,16	27	28	29	25	840		
Q13	3$\bar{Q}$		tTHL	23,22,12,13,38,39,15,16	27	28	9	20	680		
P14	4Q		tTLH	23,22,12,33,27,28,15,16	38	39	40	25	840		
Q14	4$\bar{Q}$		tTHL	23,22,12,33,27,28,15,16	38	39	20	20	680		
P15	5Q		tTLH	23,22,17,33,27,28,38,39	15	16	17	25	840		
Q15	5$\bar{Q}$	Transition time, CL = 150 pF, VDD = 5V	tTHL	23,22,17,33,27,28,38,39	15	16	36	20	680	nS	14,15

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Check the source to verify that this is the current version before use.

* TABLE XV. Transition time (tTLH and tTHL) test - Continued.

Test cond	Parameter		Connect A to pin	B to pin	C to pin	D to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
Q16	1Q	Transition time, CL = 150 pF, VDD = 5V tTHL	12,33,27,28,38,39,15,16	23	22	21	20	680	nS	14,15
P16	1$\bar{Q}$	tTLH	12,33,27,28,38,39,15,16	23	22	2	25	840		
Q17	2Q	tTHL	23,22,27,28,38,39,15,16	12	33	30	20	680		
P17	2$\bar{Q}$	tTLH	23,22,27,28,38,39,15,16	12	33	11	25	840		
Q18	3Q	tTHL	23,22,12,33,38,39,15,16	27	28	29	20	680		
P18	3$\bar{Q}$	tTLH	23,22,12,33,38,39,15,16	27	28	9	25	840		
Q19	4Q	tTHL	23,22,12,33,27,28,15,16	38	39	40	20	680		
P19	4$\bar{Q}$	tTLH	23,22,12,33,27,28,15,16	38	39	20	25	840		
Q20	5Q	tTHL	23,22,12,33,27,28,38,39	15	16	17	20	680		
P20	5$\bar{Q}$	Transition time, CL = 150 pF, VDD = 5V tTLH	23,22,12,33,27,28,38,39	15	16	38	25	840	nS	14,15

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* TABLE XVI. Propagation delay time (tPLH and tPHL) test.

Test cond	Parameter			Connect				Initial limits 25° ± 5°C ambient		Unit	Test fig
				A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1Q	Propagation delay time, CL = 50 pF, (clock to output), VDD = 5V	tPLH	12,23,27,28,38,39,15,16	23	22	21	11	350	nS	14,16
S1	1$\bar{Q}$		tPHL	12,33,27,28,38,39,15,16	23	22	2				
R2	2Q		tPLH	23,22,27,28,38,39,15,16	12	33	30				
S2	2$\bar{Q}$		tPHL	23,22,27,28,38,39,15,16	12	33	11				
R3	3Q		tPLH	23,22,12,33,38,39,15,16	27	28	29				
S3	3$\bar{Q}$		tPHL	23,22,12,33,38,39,15,16	27	28	9				
R4	4Q		tPLH	23,22,12,33,27,28,15,16	38	39	40				
S4	4$\bar{Q}$		tPHL	23,22,12,33,27,28,15,16	38	39	20				
R5	5Q		tPLH	23,22,12,33,27,28,38,39	15	16	17				
S5	5$\bar{Q}$		tPHL	23,22,12,33,27,28,38,39	15	16	36				
S6	1Q		tPHL	12,33,27,28,38,39,15,16	23	22	21				
R6	1$\bar{Q}$		tPLH	12,33,27,28,38,39,15,16	23	22	2				
S7	2Q		tPHL	23,22,27,28,38,39,15,16	12	33	30				
R7	2$\bar{Q}$		tPLH	23,22,27,28,38,39,15,16	12	33	11				
S8	3Q	Propagation delay time, CL = 50 pF, (clock to output), VDD = 5V	tPHL	23,22,27,28,38,39,15,16	27	28	29	11	350	nS	14,16

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* TABLE XVI. Propagation delay time (tPLH and tPHL) test - Continued.

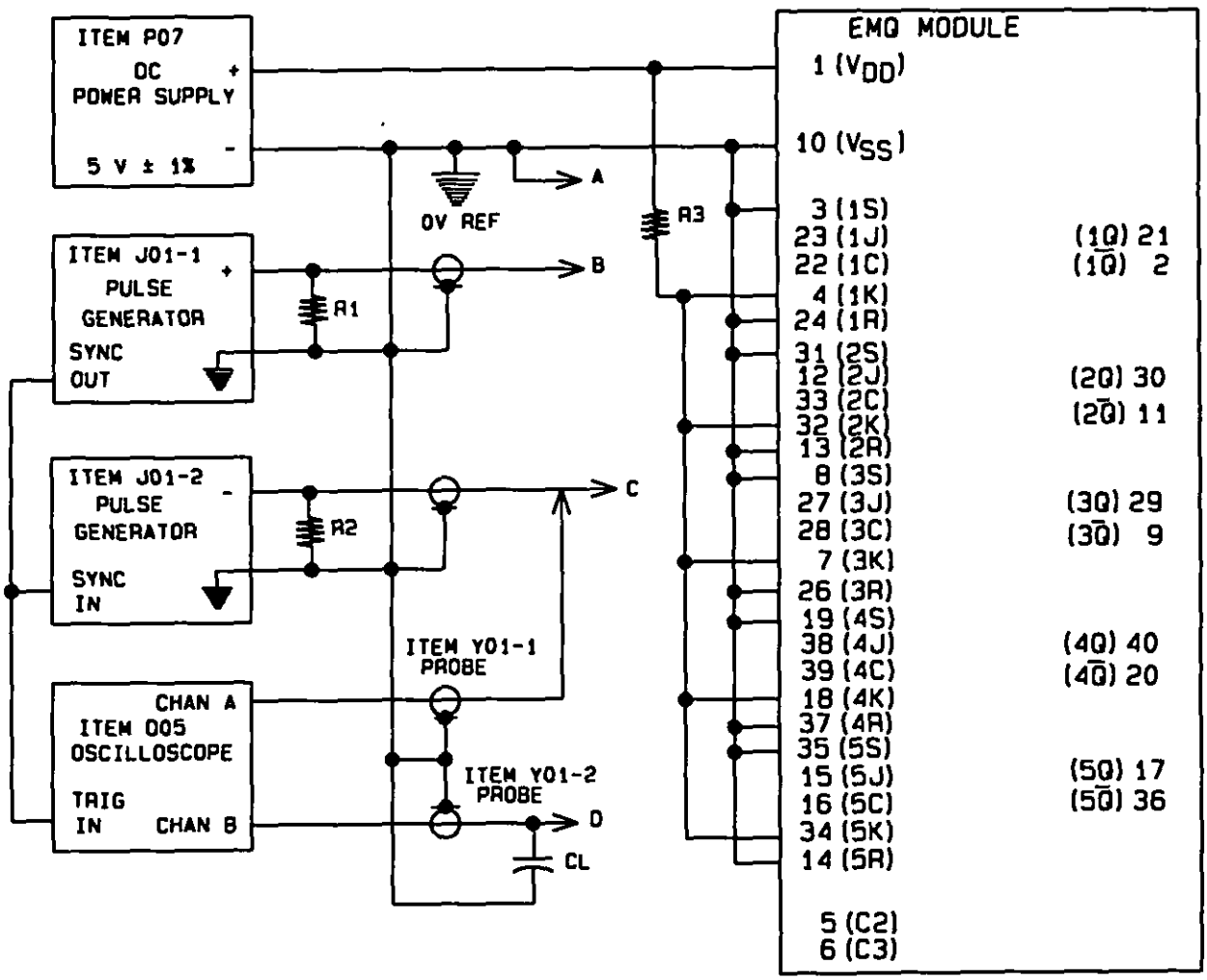
Test cond	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
R8	3Q̄ Propagation delay time, CL = 50 pF, (clock to output), VDD = 5V	tPLH	23,22,12,33,38,39,15,16	27	28	9	11	350	nS	14,16
S9	4Q	tPHL	23,22,12,33,27,28,15,16	38	39	40				
R9	4Q̄	tPLH	23,22,12,33,27,28,15,16	38	39	20				
S10	5Q	tPHL	23,22,12,33,27,28,38,39	15	16	17				
R10	5Q̄ Propagation delay time, CL = 50 pF, (clock to output), VDD = 5V	tPLH	23,22,12,33,27,28,38,39	15	16	36	11	350		
R11	1Q Propagation delay time, CL = 150 pF, (clock to output), VDD = 5V	tPLH	12,33,27,28,38,39,15,16	23	22	21	17	550		
S11	1Q̄	tPHL	12,33,27,28,38,39,15,16	23	22	2				
R12	2Q	tPLH	23,22,27,28,38,39,15,16	12	33	30				
S12	2Q̄	tPHL	23,22,27,28,38,39,15,16	12	33	11				
R13	3Q	tPLH	23,22,12,33,38,39,15,16	27	28	29				
S13	3Q̄	tPHL	23,22,12,33,38,39,15,16	27	28	9				
R14	4Q	tPLH	23,22,12,33,27,28,15,16	38	39	40				
S14	4Q̄	tPHL	23,22,12,33,27,28,15,16	38	39	20				
R15	5Q	tPLH	23,22,12,33,27,28,38,39	15	16	17				
S15	5Q̄	tPHL	23,22,12,33,27,28,38,39	15	16	36				
S16	1Q Propagation delay time, CL = 150 pF, (clock to output), VDD = 5V	tPHL	12,33,27,28,38,39,15,16	23	22	21	17	550	nS	14,16

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* TABLE XVI. Propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter		Connect A to pin	B to pin	C to pin	D to pin	Initial limits 25° ± 5°C ambient Min	Max	Unit	Test fig
R16	$1\bar{Q}$ Propagation delay time, CL = 150 pF, (clock to output), VDD = 5V	tPLH	12,33,27,28,38,39,15,16	23	22	2	17	550	nS	14,16
S17	2Q	tPHL	23,22,27,28,38,39,15,16	12	33	30				
R17	$2\bar{Q}$	tPLH	23,22,27,28,38,39,15,16	12	33	11				
S18	3Q	tPHL	23,22,12,33,38,39,15,16	27	28	29				
R18	$3\bar{Q}$	tPLH	23,22,12,33,38,39,15,16	27	28	9				
S19	4Q	tPHL	23,22,12,33,27,28,15,16	38	39	40				
R19	$4\bar{Q}$	tPLH	23,22,12,33,27,28,15,16	38	39	20				
S20	5Q	tPHL	23,22,12,33,27,28,38,39	15	16	17				
R20	$5\bar{Q}$ Propagation delay time, CL = 150 pF, (clock to output), VDD = 5V	tPLH	23,22,12,33,27,28,38,39	15	16	36	17	550	nS	14,16

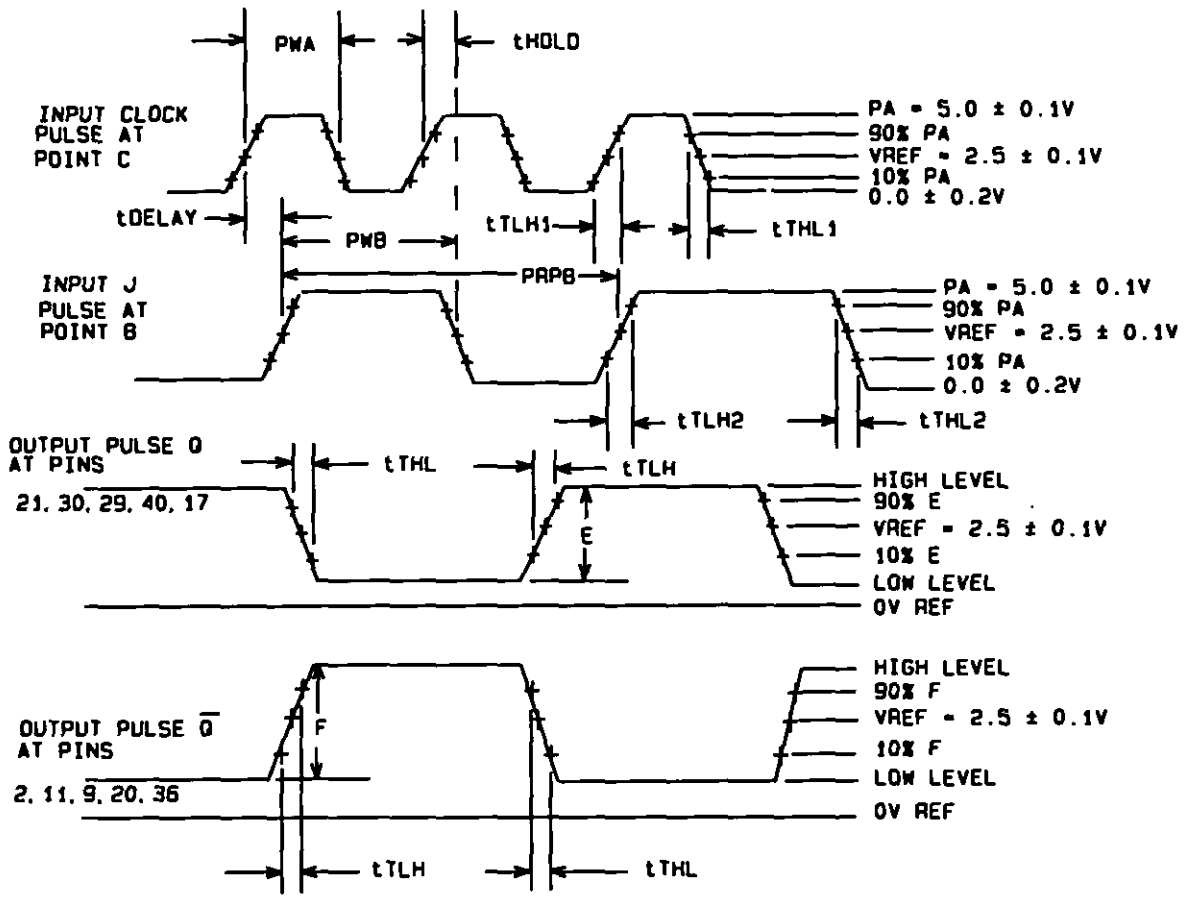
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NOTES:
1. R1 = R2 = 51.1 Ohms ± 5 percent, 1/2 W.
2. CL = 50 ± 5 pF (including probe and parasitic capacitance) for test conditions P1-P10, Q1-Q10, R1-R10, and S1-S10.
3. CL = 150 ± 15 pF (including probe and parasitic capacitance) for test conditions P11-P20, Q11-Q20, R11-R20, and S11-S20.
4. R3 = 1.5 kOhms ± 1 percent, 1/4 W.

FIGURE 14. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test circuit for tables XV and XVI.

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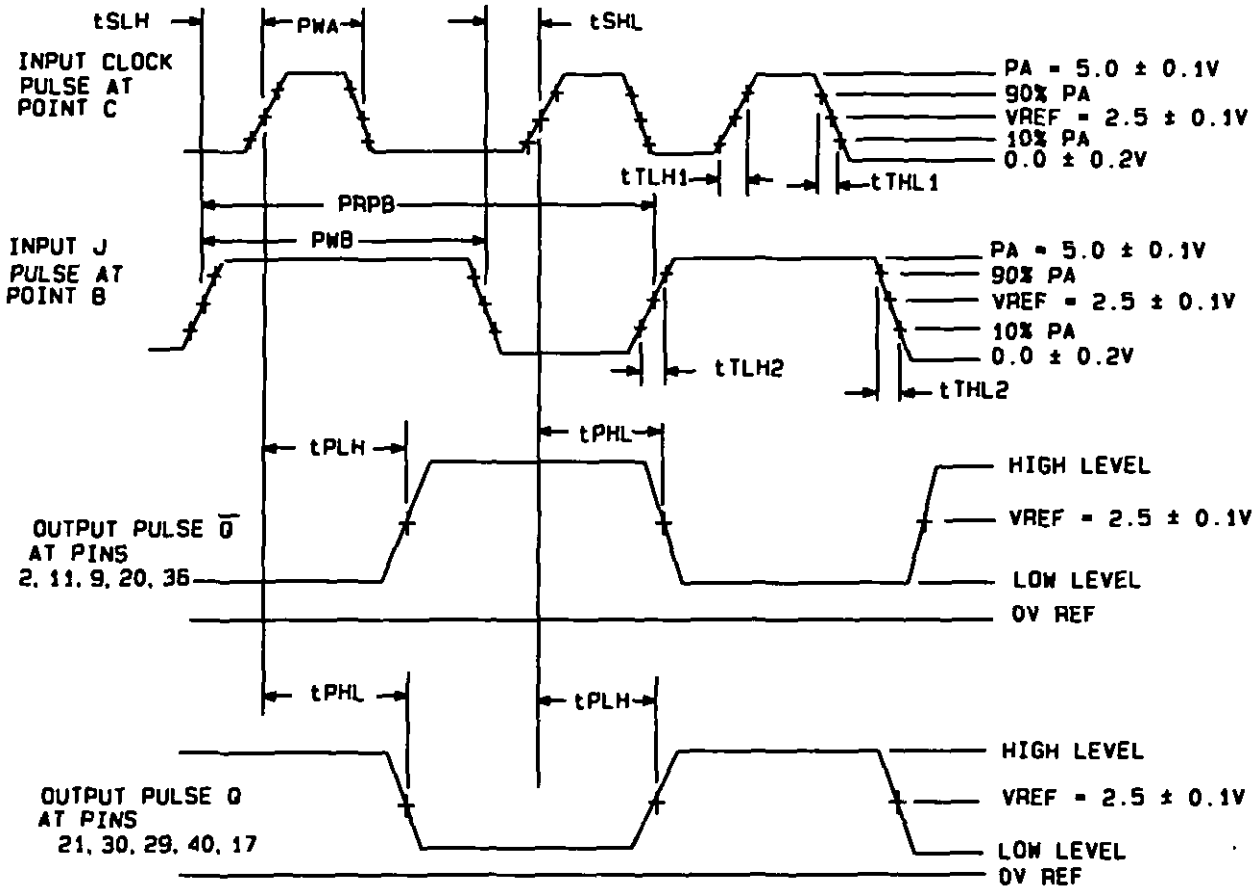


NOTES:

1. Input pulse parameters for item J01-2:
   Put item J01-2 in the double pulse mode.
   tTLH1 = tTHL1 = 10 ± 2 nS.
   PWA = 1.0 ± 0.1 µS.
2. Input pulse parameters for item J01-1:
   tTLH2 = tTHL2 = 10 ± 2 nS.
   PWB = 1.0 ± 0.1 µS.
   PRPB = 4.0 ± 0.4 µS.
3. tHOLD = 0 +2 nS, -0 nS.
   tDELAY = 1.0 ± 0.1 µS.
4. Verify that all outputs have a high level ≥ 4.6V and a low level ≤ 0.4V before making measurements.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to ground.

FIGURE 15. Transition time (tTLH and tTHL) test circuit waveforms for table XV.

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NOTES:

1. Input pulse parameters for item J01-2:
   Put item J01-2 in the double pulse mode.
   tTLH1 = tTHL1 = 10 ± 2 nS.
   PWA = 1.0 ± 0.1 µS.
2. Input pulse parameters for item J01-1:
   tTLH2 = tTHL2 = 10 ± 2 nS.
   PWB = 2.0 ± 0.2 µS.
   PRPB = 4.0 ± 0.4 µS.
3. tSLH = tSHL = 200 nS ± 5 percent.
4. Verify that all outputs have a high level ≥ 4.6V and a low level ≤ 0.4V before making measurements.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to ground.

FIGURE 16. Propagation delay time (tPLH and tPHL) test circuit waveforms for table XVI.

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* TABLE XVII. Propagation delay time (tPLH and tPHL) test.

Test cond	Parameter		Connect A to pin	B to pin	C to pin	D to pin	E to pin	Initial limits 25° ± 5°C ambient Min	Max	Unit	Test fig
T1	1Q Propagation delay time, CL = 50 pF, (set to output), VDD = 5V	tPLH	31,13,8,26,19,37,35,14	3	24	B	21	11	350	nS	17,18
U1	1$\bar{Q}$	tPHL	31,13,8,26,19,37,35,14	3	24		2				
T2	2Q	tPLH	3,24,8,26,19,37,35,14	31	13		30				
U2	2$\bar{Q}$	tPHL	3,24,8,26,19,37,35,14	31	13		11				
T3	3Q	tPLH	3,24,31,13,19,37,35,14	8	26		29				
U3	3$\bar{Q}$	tPHL	3,24,31,13,19,37,35,14	8	26		9				
T4	4Q	tPLH	3,24,31,13,8,26,35,14	19	37		40				
U4	4$\bar{Q}$	tPHL	3,24,31,13,8,26,35,14	19	37		20				
T5	5Q	tPLH	3,24,31,13,8,26,19,37	35	14		17				
U5	5$\bar{Q}$ Propagation delay time, CL = 50 pF (set to output), VDD = 5V	tPHL	3,24,31,13,8,26,19,37	35	14	B	38	11	350		
U6	1Q Propagation delay time, CL = 50 pF (reset to output), VDD = 5V	tPHL	31,13,8,26,19,37,35,14	3	24	C	21	14	450		
T6	1$\bar{Q}$	tPLH	31,13,8,26,19,37,35,14	3	24		2				
U7	2Q	tPHL	3,24,8,26,19,37,35,14	31	13		30				
T7	2$\bar{Q}$	tPLH	3,24,8,26,19,37,35,14	31	13		11				
U8	3Q Propagation delay time, CL = 50 pF, (reset to output), VDD = 5V	tPHL	3,24,31,13,19,37,35,14	8	26	C	29	14	450	nS	17,18

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* TABLE XVII. Propagation delay time (tPLH and tPHL) test - Continued.

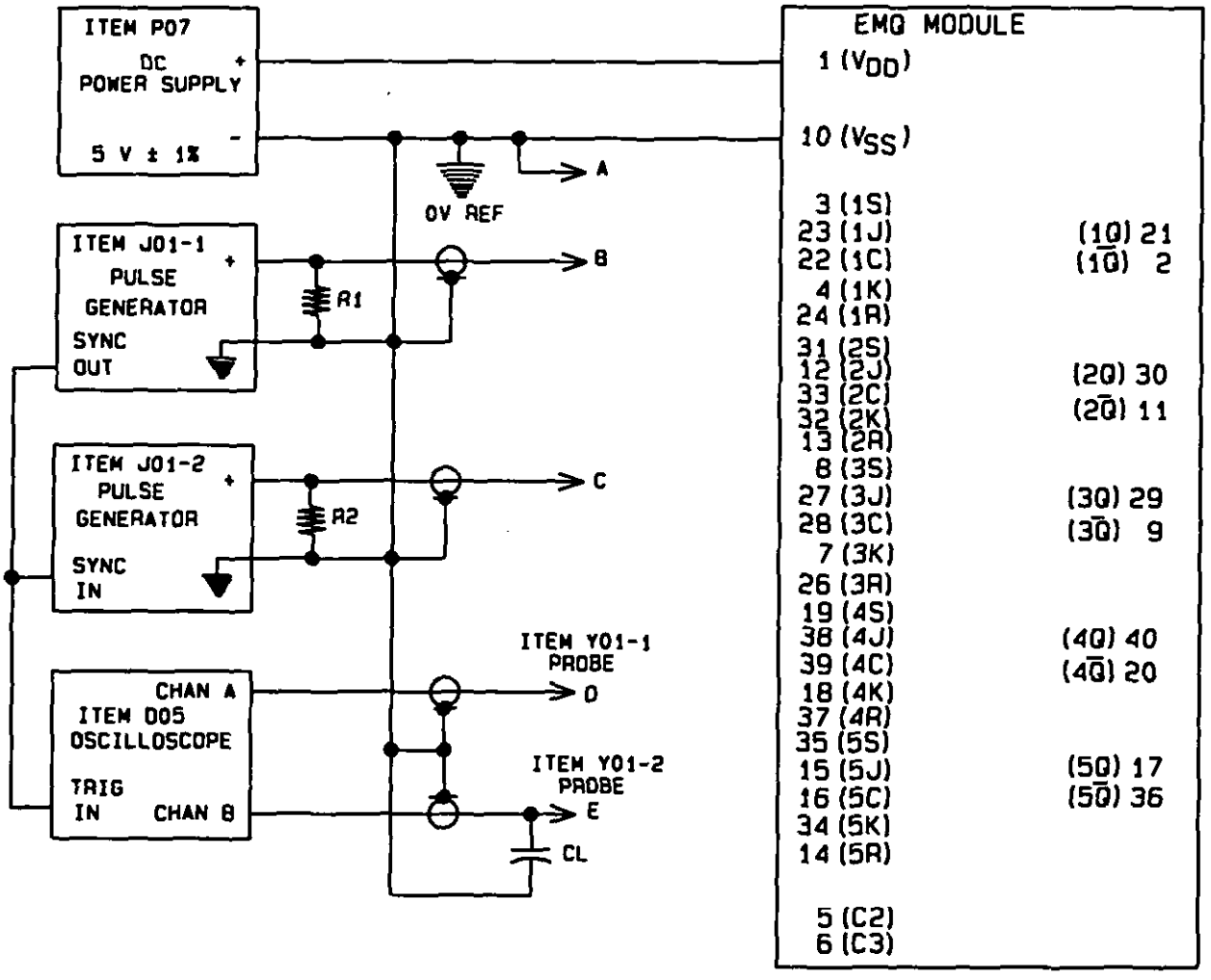
Test cond	Parameter		Connect A to pin	B to pin	C to pin	D to pin	E to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
T8	3$\bar{Q}$ Propagation delay time, CL = 50 pF, (reset to output), VDD = 5V	tPLH	3,24,31,13,19,37,35,14	8	26	C	9	14	450	nS	17,18
U9	4Q	tPHL	3,24,31,13,8,26,35,14	19	37		40				
T9	4$\bar{Q}$	tPLH	3,24,31,13,8,26,35,14	19	37		20				
U10	5Q	tPHL	3,24,31,13,8,26,19,37	35	14		17				
T10	5$\bar{Q}$ Propagation delay time, CL = 50 pF, (reset to output), VDD = 5V	tPLH	3,24,31,13,8,26,19,37	35	14	C	36	14	450		
T11	1Q Propagation delay time, CL = 150 pF, (set to output), VDD = 5V	tPLH	31,13,8,26,19,37,35,14	3	24	B	21	17	550		
U11	1$\bar{Q}$	tPHL	31,13,8,26,19,37,35,14	3	24		2				
T12	2Q	tPLH	3,24,8,26,19,37,35,14	31	13		30				
U12	2$\bar{Q}$	tPHL	3,24,8,26,19,37,35,14	31	13		11				
T13	3Q	tPLH	3,24,31,13,19,37,35,14	8	26		29				
U13	3$\bar{Q}$	tPHL	3,24,31,13,19,37,35,14	8	26		9				
T14	4Q	tPLH	3,24,31,13,8,26,35,14	19	37		40				
U14	4$\bar{Q}$	tPHL	3,24,31,13,8,26,35,14	19	37		20				
T15	5Q	tPLH	3,24,31,13,8,26,19,37	35	14		17				
U15	5$\bar{Q}$ Propagation delay time, CL = 150 pF (set to output), VDD = 5V	tPHL	3,24,31,13,8,26,19,37	35	14	B	36	17	550	nS	17,18

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* TABLE XVII. Propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter		Connect A to pin	B to pin	C to pin	D to pin	E to pin	Initial limits 25° ± 5°C ambient Min	Max	Unit	Test fig
U16	1Q Propagation delay time, CL = 150 pF, (reset to output), VDD = 5V	tPHL	31,13,8,26,19,37,35,14	3	24	C	21	20	650	nS	17,18
T16	1$\bar{Q}$	tPLH	31,13,8,26,19,37,35,14	3	24		2				
U17	2Q	tPHL	3,24,8,26,19,37,35,14	31	13		30				
T17	2$\bar{Q}$	tPLH	3,24,8,26,19,37,35,14	31	13		11				
U18	3Q	tPHL	3,24,31,13,19,37,35,14	8	26		29				
T18	3$\bar{Q}$	tPLH	3,24,31,13,19,37,35,14	8	26		9				
U19	4Q	tPHL	3,24,31,13,8,26,35,14	19	37		40				
T19	4$\bar{Q}$	tPLH	3,24,31,13,8,26,35,14	19	37		20				
U20	5Q	tPHL	3,24,31,13,8,26,19,37	35	14		17				
T20	5$\bar{Q}$ Propagation delay time, CL = 150 pF, (reset to output), VDD = 5V	tPLH	3,24,31,13,8,26,19,37	35	14	C	38	20	650	nS	17,18

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NOTES:
1. R1 = R2 = 51.1 Ohms ± 5 percent, 1/2 W.
2. CL = 50 ± 5 pF (including probe and parasitic capacitance) for test conditions T1-T10 and U1-U10.
3. CL = 150 ± 15 pF (including probe and parasitic capacitance) for test conditions T11-T20 and U11-U20.

FIGURE 17. Propagation delay time (tPLH and tPHL) test circuit for table XVII.

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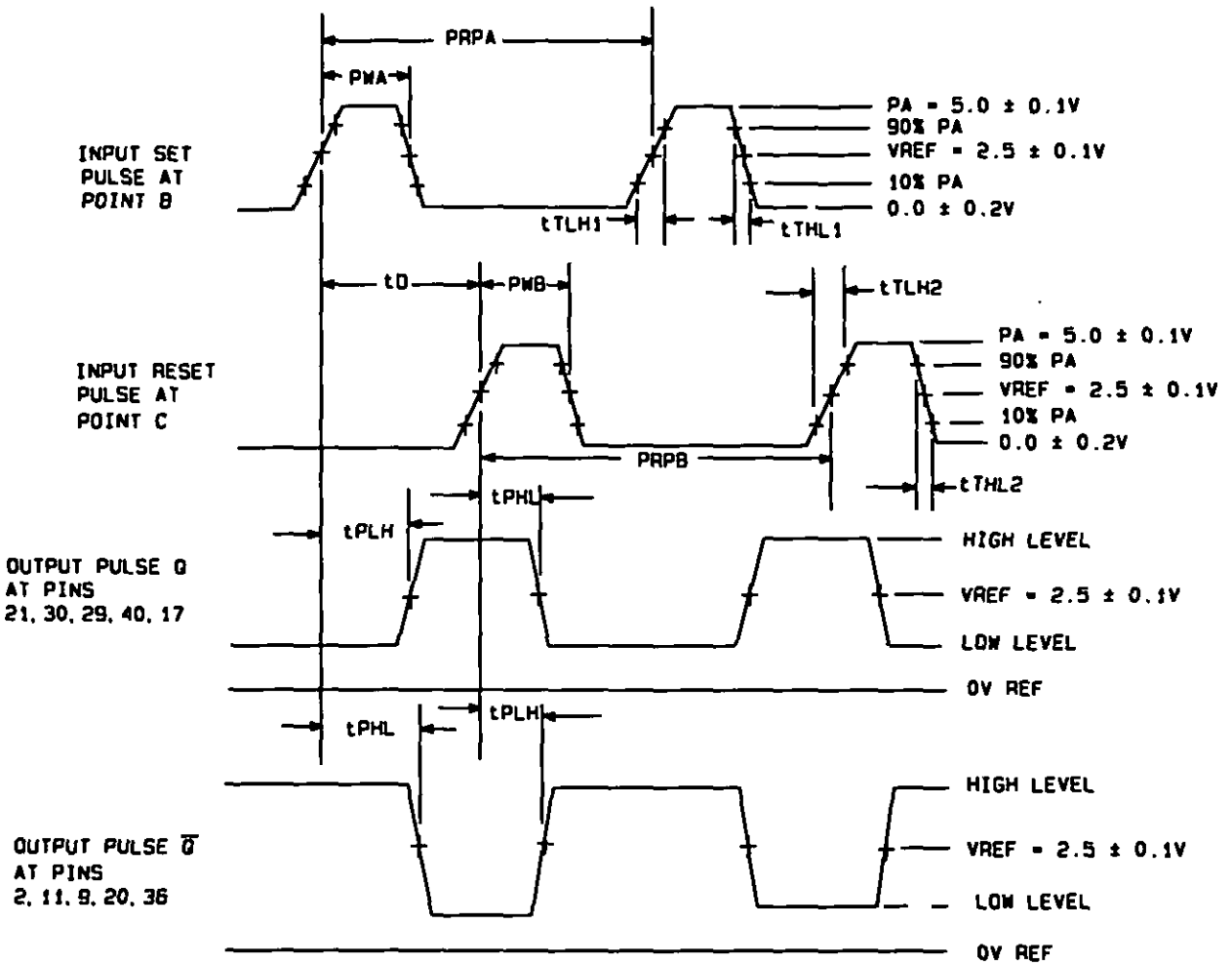


FIGURE 18. Propagation delay time (tPLH and tPHL) test circuit waveforms for table XVII.

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NOTES:
1. Input pulse parameters for item J01-1:
   PRPA = 4.0 ± 0.4 μS.
   tTLH1 = tTHL1 = 10 ± 2 nS.
   PWA = 250 ± 20 nS.
2. Input pulse parameters for item J01-2:
   PRPB = 4.0 ± 0.4 μS.
   tTLH2 = tTHL2 = 10 ± 2 nS.
   PWB = 250 ± 20 nS.
3. tD = 2.0 ± 0.2 μS.
4. Verify that all outputs have a high level ≥ 4.6V and a low level ≤ 0.4V before making measurements.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
6. All voltages are referenced to ground.

FIGURE 18. Propagation delay time (tPLH and tPHL) test circuit waveforms for table XVII - Continued.

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* TABLE XVIII. Transition time (tTLH and tTHL) test.

Test cond	Parameter		Connect A to pin	B to pin	C to pin	D to pin	Initial limits 25° ± 5°C ambient Min	Max	Unit	Test fig
V1	1Q	Transition time, CL = 50 pF, VDD = 10V tTLH	12,33,27,28,38,39,15,16	23	22	21	5	180	nS	19,20
W1	1$\bar{Q}$	tTHL	12,33,27,28,38,39,15,16	23	22	2	3	100		
V2	2Q	tTLH	23,22,27,28,38,39,15,16	12	33	30	5	180		
W2	2$\bar{Q}$	tTHL	23,22,27,28,38,39,15,16	12	33	11	3	100		
V3	3Q	tTLH	23,22,12,33,38,39,15,16	27	28	29	5	180		
W3	3$\bar{Q}$	tTHL	23,22,12,33,38,39,15,16	27	28	9	3	100		
V4	4Q	tTLH	23,22,12,33,27,28,15,16	38	39	40	5	180		
W4	4$\bar{Q}$	tTHL	23,22,12,33,27,28,15,16	38	39	20	3	100		
V5	5Q	tTLH	23,22,12,33,27,28,38,39	15	16	17	5	180		
W5	5$\bar{Q}$	tTHL	23,22,12,33,27,28,38,39	15	16	36	3	100		
W6	1Q	tTHL	12,33,27,28,38,39,15,16	23	22	21	3	100		
V6	1$\bar{Q}$	tTLH	12,33,27,28,38,39,15,16	23	22	2	5	180		
W7	2Q	tTHL	23,22,27,28,38,39,15,16	12	33	30	3	100		
V7	2$\bar{Q}$	tTLH	23,22,27,28,38,39,15,16	12	33	11	5	180		
W8	3Q	tTHL	23,22,12,33,38,39,15,16	27	28	29	3	100		
V8	3$\bar{Q}$	Transition time, CL = 50 pF, VDD = 10V tTLH	23,22,12,33,38,39,15,16	27	28	9	5	180	nS	19,20

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* TABLE XVIII. Transition time (tTLH and tTHL) test - Continued.

Test cond	Parameter		Connect				Initial limits 25° ± 5°C ambient		Unit	Test fig
			A to pin	B to pin	C to pin	D to pin	Min	Max		
W9	4Q	Transition time, CL = 50 pF, VDD = 10V tTHL	23,22,12,33,27,28,15,16	38	39	40	3	100	nS	19,20
V9	4$\bar{Q}$	tTLH	23,22,12,33,27,28,15,16	38	39	20	5	180		
W10	5Q	tTHL	23,22,12,33,27,28,38,39	15	16	17	3	100		
V10	5$\bar{Q}$	Transition time, CL = 50 pF, VDD = 10V tTLH	23,22,12,33,27,28,38,39	15	16	38	5	180	nS	19,20

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* TABLE XIX. Propagation delay time (tPLH and tPHL) test.

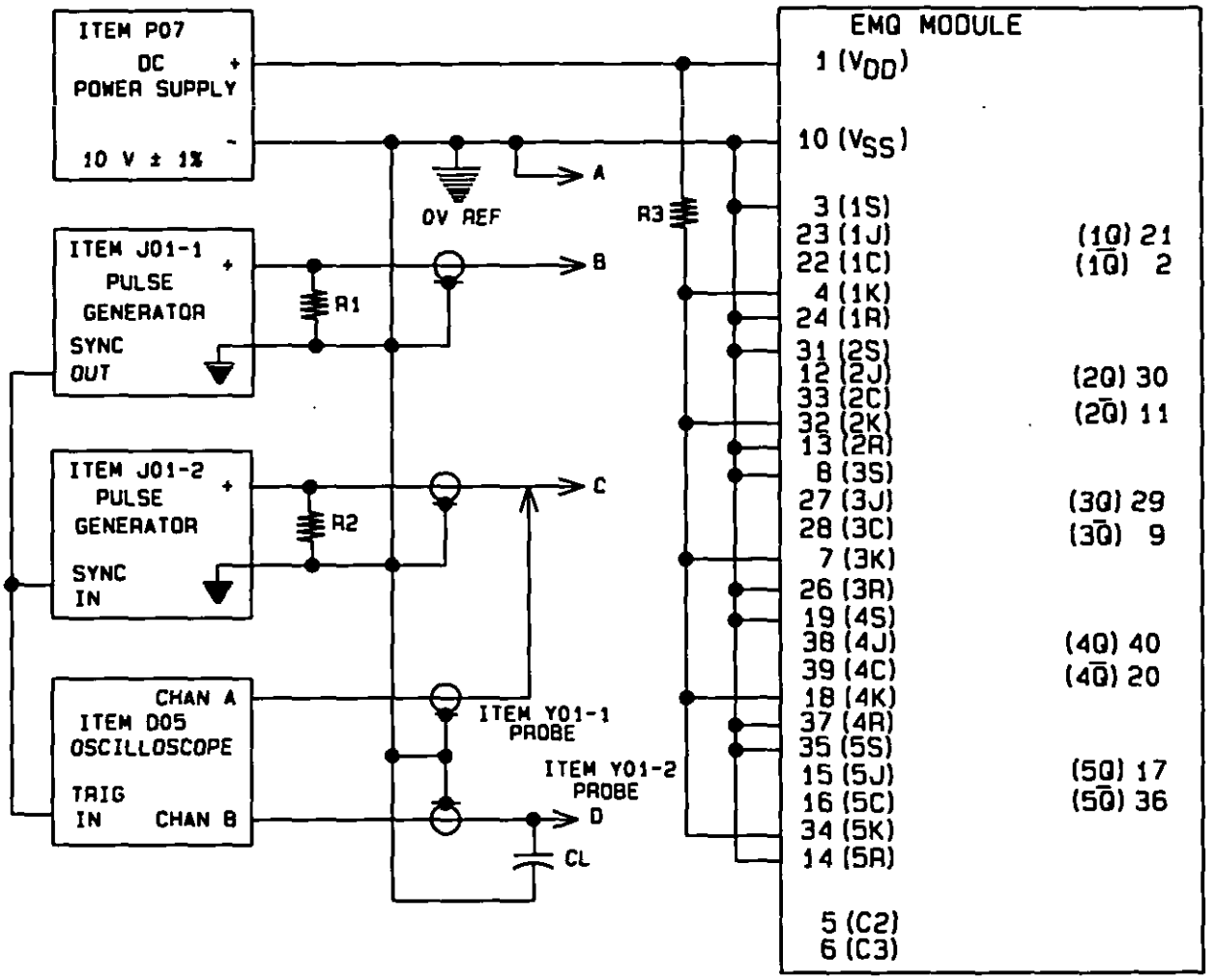
Test cond	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
X1	1Q	Propagation delay time, CL = 50 pF, (clock to output), VDD = 10V tPLH	12,33,27,28,38,39,15,16	23	22	21	5	150	nS	19,21
Y1	1$\bar{Q}$	tPHL	12,33,27,28,38,39,15,16	23	22	2				
X2	2Q	tPLH	23,22,27,28,38,39,15,16	12	33	30				
Y2	2$\bar{Q}$	tPHL	23,22,27,28,38,39,15,16	12	33	11				
X3	3Q	tPLH	23,22,12,33,38,39,15,16	27	28	29				
Y3	3$\bar{Q}$	tPHL	23,22,12,33,38,39,15,16	27	28	9				
X4	4Q	tPLH	23,22,12,33,27,28,15,16	38	39	40				
Y4	4$\bar{Q}$	tPHL	23,22,12,33,27,28,15,16	38	39	20				
X5	5Q	tPLH	23,22,12,33,27,28,38,39	15	16	17				
Y5	5$\bar{Q}$	tPHL	23,22,12,33,27,28,38,39	15	16	36				
Y6	1Q	tPHL	12,33,27,28,38,39,15,16	23	22	21				
X6	1$\bar{Q}$	tPLH	12,33,27,28,38,39,15,16	23	22	2				
Y7	2Q	tPHL	23,22,27,28,38,39,15,16	12	33	30				
X7	2$\bar{Q}$	tPLH	23,22,27,28,38,39,15,16	12	33	11				
Y8	3Q	tPHL	23,22,12,33,38,39,15,16	27	28	29				
X8	3$\bar{Q}$	Propagation delay time, CL = 50 pF, (clock to output), VDD = 10V tPLH	23,22,12,33,38,39,15,16	27	28	9	5	150	nS	19,21

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* TABLE XIX. Propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter			Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
Y9	4Q	Propagation delay time, CL = 50 pF, (clock to output), VDD = 10V	tPHL	23,22,12,33,27,28,15,16	38	39	40	5	150	nS	19,21
X9	$4\overline{Q}$		tPLH	23,22,12,33,27,28,15,18	38	39	20				
Y10	5Q		tPHL	23,22,12,33,27,28,38,39	15	16	17				
X10	$5\overline{Q}$	Propagation delay time, CL = 50 pF, (clock to output), VDD = 10V	tPLH	23,22,12,33,27,28,38,39	15	16	38	5	150	nS	19,21

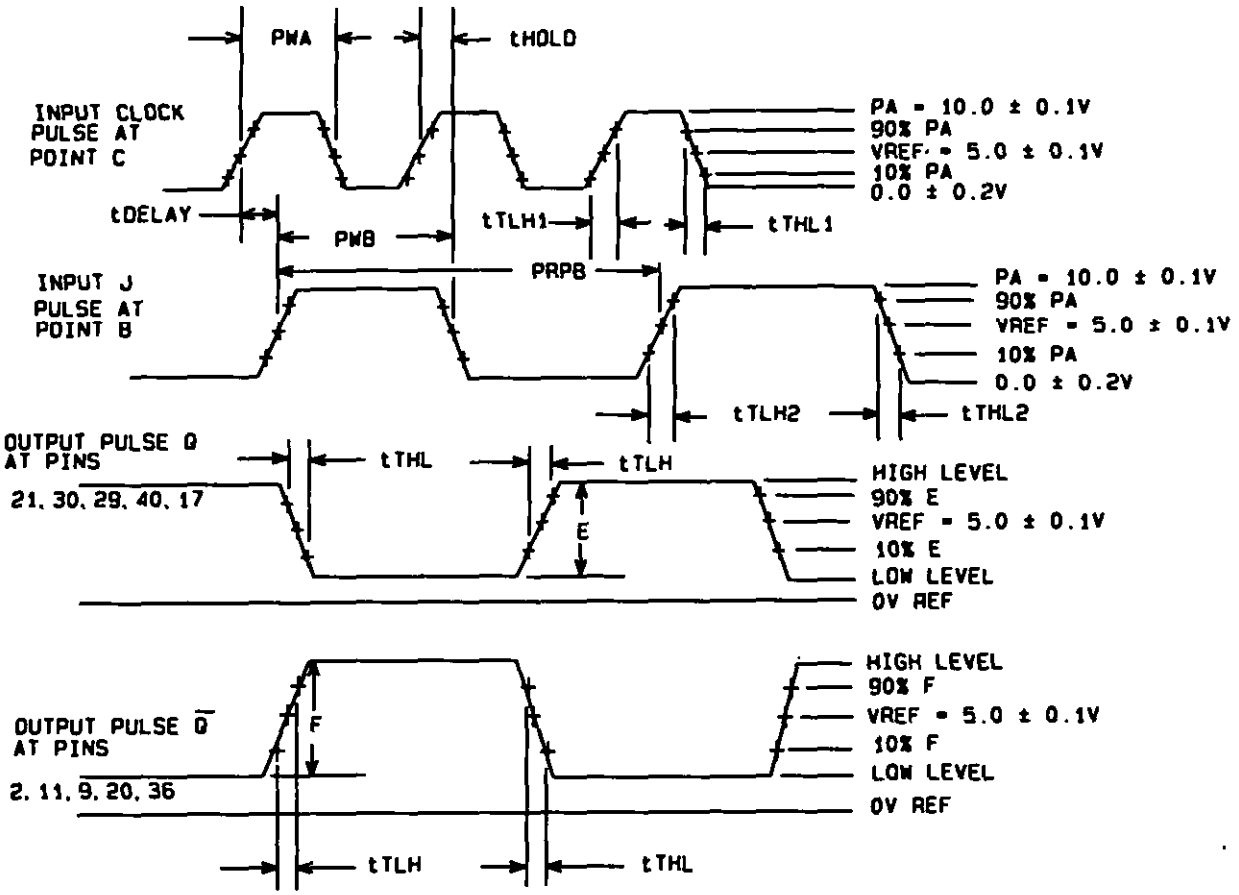
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:13Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 51.1 Ohms ± 5 percent, 2 W.
2. R3 = 1.5 kOhms ± 1 percent, 1/4 W.
3. CL = 50 ± 5 pF (including probe and parasitic capacitance).

* FIGURE 19. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test circuit for tables XVIII and XIX.

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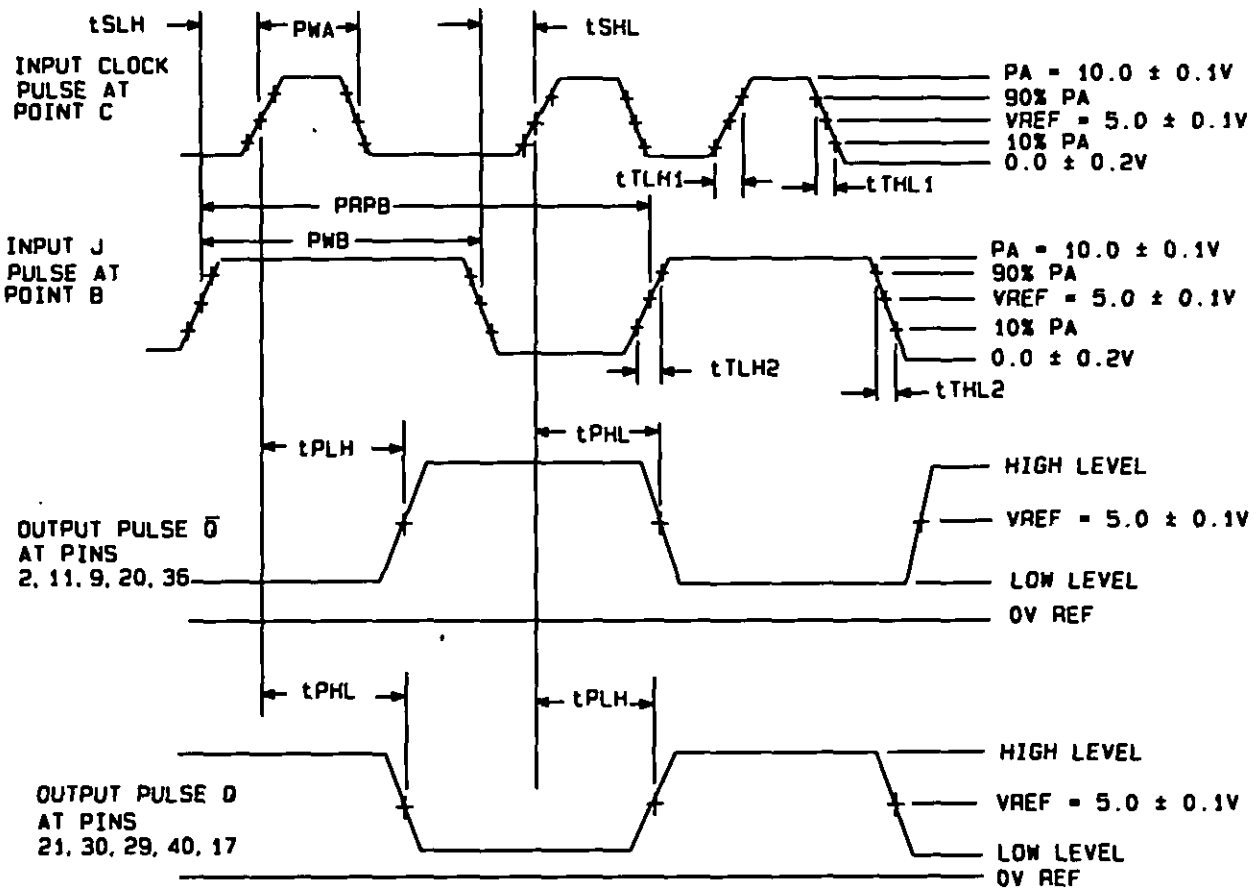


NOTES:

1. Input pulse parameters for item J01-2:
   Put item J01-2 in the double pulse mode.
   tTLH1 = tTHL1 = 10 ± 2 nS.
   PWA = 1.0 ± 0.1 μS.
2. Input pulse parameters for item J01-1:
   tTLH2 = tTHL2 = 10 ± 2 nS.
   PWB = 2.0 ± 0.2 μS.
   PRPB = 4.0 ± 0.4 μS.
3. tHOLD = 0 +2 nS, -0 nS.
   tDELAY = 1.0 ± 0.1 μS.
4. Verify that all outputs have a high level ≥ 9.5V and a low level ≤ 0.5V before making measurements.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
6. All voltages are referenced to ground.

FIGURE 20. Transition time (tTLH and tTHL) test circuit waveforms for table XVIII.

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NOTES:
1. Input pulse parameters for item J01-2:
   Put item J01-2 in the double pulse mode.
   tTLH1 = tTHL1 = 10 ± 2 nS.
   PWA = 1.0 ± 0.1 μS.
2. Input pulse parameters for item J01-1:
   tTLH2 = tTHL2 = 10 ± 2 nS.
   PWB = 2.0 ± 0.2 μS.
   PRPB = 4.0 ± 0.4 μS.
3. tSLH = tSHL = 200 nS ± 5 percent.
4. Verify that all outputs have a high level ≥ 9.5V and a low level ≤ 0.5V before making measurements.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
6. All voltages are referenced to ground.

FIGURE 21. Propagation delay time (tPLH and tPHL) test circuit waveforms for table XIX.

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* TABLE XX. Propagation delay time (tPLH and tPHL) test.

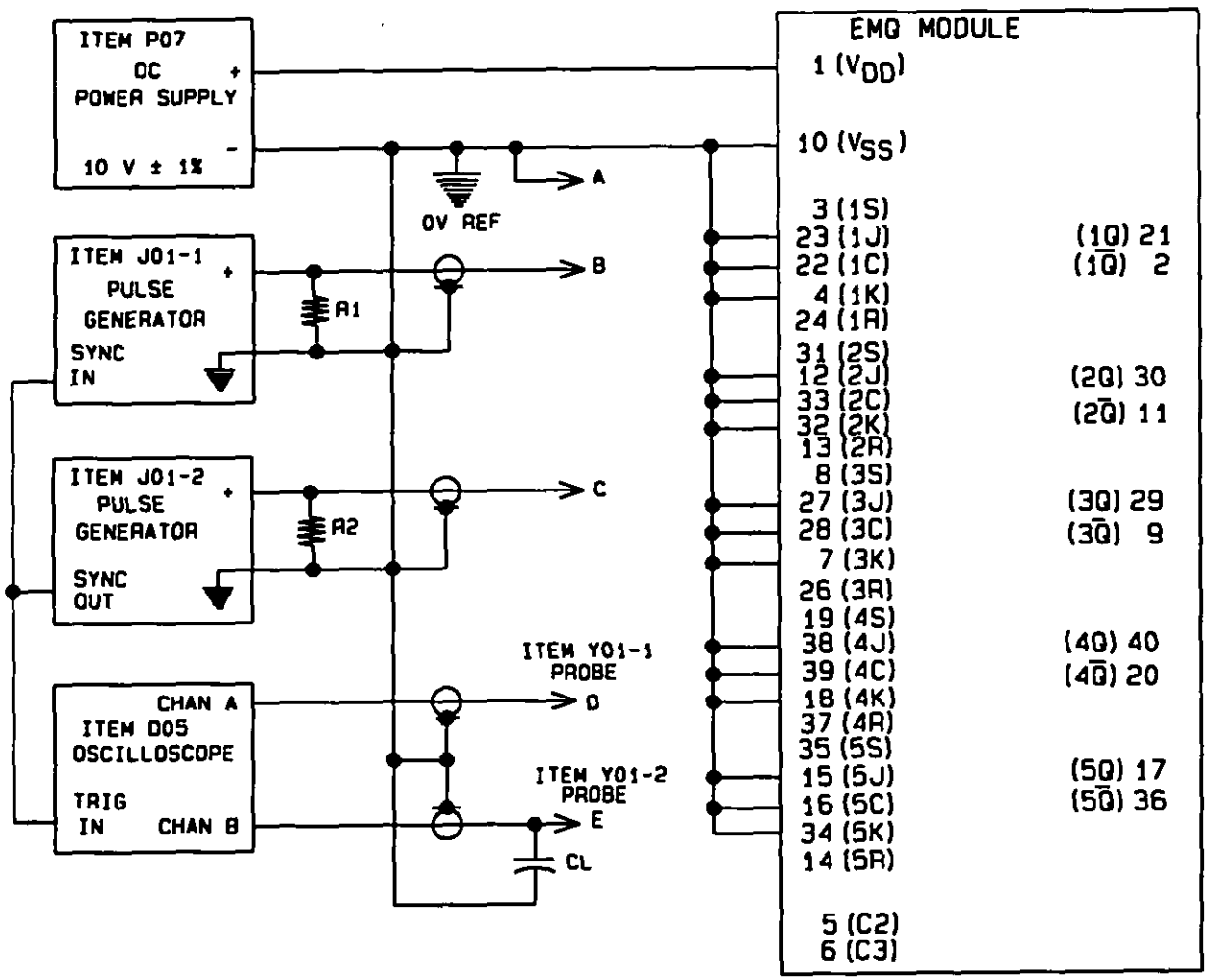
Test cond	Parameter		Connect A to pin	B to pin	C to pin	D to pin	E to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
Z1	1Q Propagation delay time, CL = 50 pF (set to output), VDD = 10V	tPLH	31,13,8,26,19,37,35,14	3	24	B	21	5	150	nS	22,23
AA1	1$\bar{Q}$	tPHL	31,13,8,26,19,37,35,14	3	24		2				
Z2	2Q	tPLH	3,24,8,26,19,37,35,14	31	13		30				
AA2	2$\bar{Q}$	tPHL	3,24,8,26,19,37,35,14	31	13		11				
Z3	3Q	tPLH	3,24,31,13,19,37,35,14	8	26		29				
AA3	3$\bar{Q}$	tPHL	3,24,31,13,19,37,35,14	8	26		9				
Z4	4Q	tPLH	3,24,31,13,8,26,35,14	19	37		40				
AA4	4$\bar{Q}$	tPHL	3,24,31,13,8,26,35,14	19	37		20				
Z5	5Q	tPLH	3,24,31,13,8,26,19,37	35	14		17				
AA5	5$\bar{Q}$ Propagation delay time, CL = 50 pF (set to output), VDD = 10V	tPHL	3,24,31,13,8,26,19,37	35	14	B	36	5	150		
AA6	1Q Propagation delay time, CL = 50 pF (reset to output), VDD = 10V	tPHL	31,13,8,26,19,37,35,14	3	24	C	21	6	200		
Z6	1$\bar{Q}$	tPLH	31,13,8,26,19,37,35,14	3	24		2				
AA7	2Q	tPHL	3,24,8,26,19,37,35,14	31	13		30				
Z7	2$\bar{Q}$	tPLH	3,24,8,26,19,37,35,14	31	13		11				
AA8	3Q	tPHL	3,24,31,13,19,37,35,14	8	26		29				
Z8	3$\bar{Q}$ Propagation delay time, CL = 50 pF (reset to output), VDD = 10V	tPLH	3,24,31,13,19,37,35,14	8	26	C	9	6	200	nS	22,23

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* TABLE XX. *Propagation delay time (tPLH and tPHL) test* - Continued.

Test cond	Parameter			Connect						Initial limits 25° ± 5°C ambient		Unit	Test fig
				A to pin	B to pin	C to pin	D to pin	E to pin		Min	Max		
AA9	4Q	Propagation delay time, CL = 50 pF (*reset to output*), VDD = 10V	tPHL	3,24,31,13,8,28,35,14	19	37	C	40		8	200	nS	22,23
Z9	4$\overline{Q}$		tPLH	3,24,31,13,8,28,35,14	18	37		20					
AA10	5Q		tPHL	3,24,31,13,8,28,18,37	35	14		17					
Z10	5$\overline{Q}$	Propagation delay time, CL = 50 pF (*reset to output*), VDD = 10V	tPLH	3,24,31,13,8,28,19,37	35	14	C	36		8	200	nS	22,23

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NOTES:
1. R1 = R2 = 51.1 Ohms ± 5 percent, 2 W.
2. CL = 50 ± 5 pF (including probe and parasitic capacitance).

FIGURE 22. Propagation delay time (tPLH and tPHL) test circuit for table XX.

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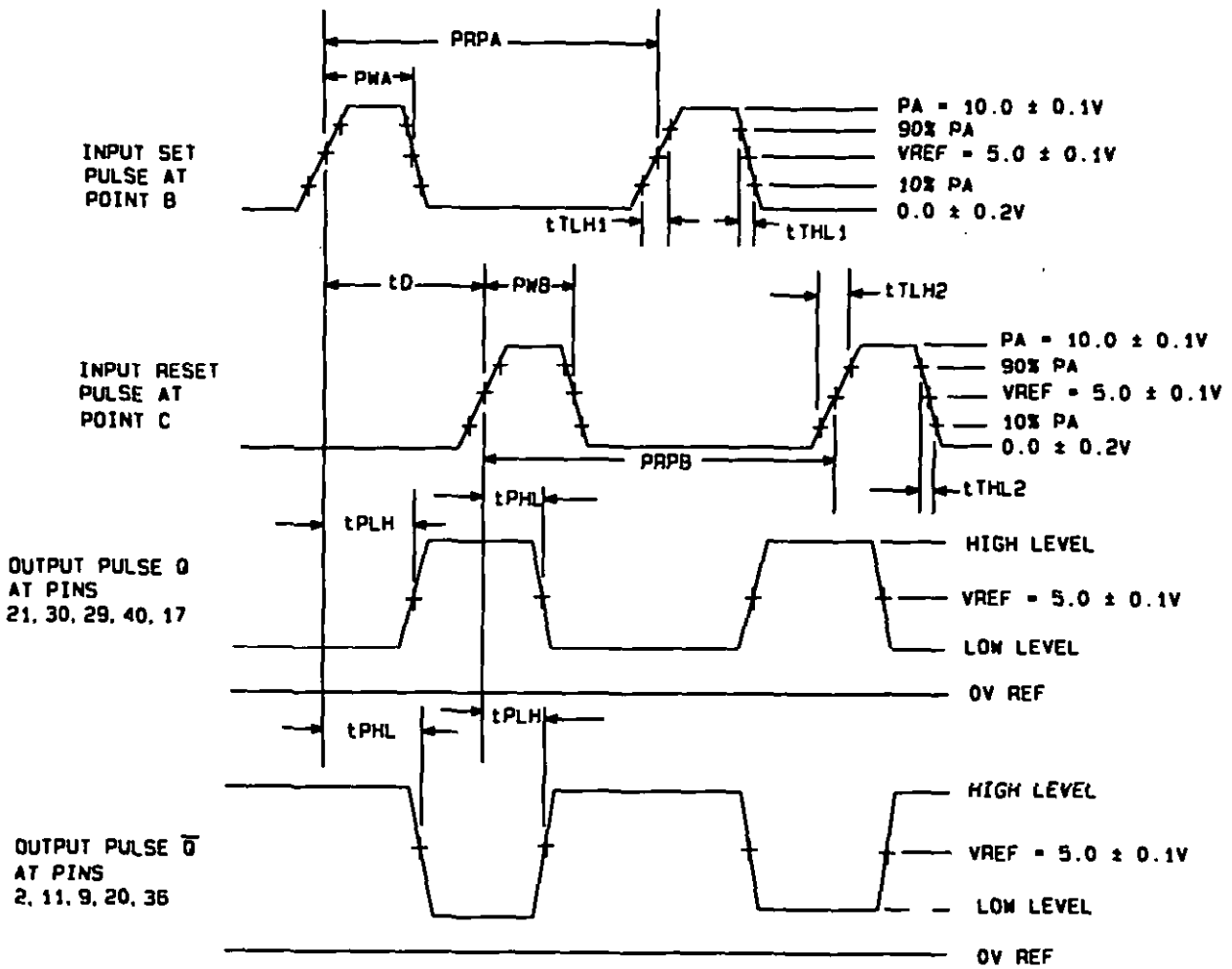


FIGURE 23. Propagation delay time (tTLH and tTHL) test circuit waveforms for table XX.

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NOTES:
1. Input pulse parameters for item J01-1:
   PRPA = 4.0 ± 0.4 μS.
   tTLH1 = tTHL1 = 10 ± 2 nS.
   PWA = 250 ± 20 nS.
2. Input pulse parameters for item J01-2:
   PRPB = 4.0 ± 0.4 μS.
   tTLH2 = tTHL2 = 10 ± 2 nS.
   PWB = 250 ± 20 nS.
3. tD = 2.0 ± 0.2 μS.
4. Verify that all outputs have a high level ≥ 9.6V and a low level ≤ 0.5V before making measurements.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
6. All voltages are referenced to ground.

FIGURE 23. Propagation delay time (tTLH and tTHL) test circuit waveforms for table XX - Continued.

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* TABLE XXI. Transition time (tTLH and tTHL) test.

Test cond	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
BB1	1Q	tTLH	12,33,27,28,38,39,15,16	23	22	21	4	130	nS	24,25
CC1	1$\bar{Q}$	tTHL	12,33,27,28,38,39,15,16	23	22	2	2	80		
BB2	2Q	tTLH	23,22,27,28,38,39,15,16	12	33	30	4	130		
CC2	2$\bar{Q}$	tTHL	23,22,27,28,38,39,15,16	12	33	11	2	80		
BB3	3Q	tTLH	23,22,12,33,38,39,15,16	27	28	29	4	130		
CC3	3$\bar{Q}$	tTHL	23,22,12,33,38,39,15,16	27	28	9	2	80		
BB4	4Q	tTLH	23,22,12,33,27,28,15,16	38	39	40	4	130		
CC4	4$\bar{Q}$	tTHL	23,22,12,33,27,28,15,16	38	39	20	2	80		
BB5	5Q	tTLH	23,22,12,33,27,28,38,39	15	16	17	4	130		
CC5	5$\bar{Q}$	tTHL	23,22,12,33,27,28,38,39	15	16	36	2	80		
CC6	1Q	tTHL	12,33,27,28,38,39,15,16	23	22	21	2	80		
BB6	1$\bar{Q}$	tTLH	12,33,27,28,38,39,15,16	23	22	2	4	130		
CC7	2Q	tTHL	23,22,27,28,38,39,15,16	12	33	30	2	80		
BB7	2$\bar{Q}$	tTLH	23,22,27,28,38,39,15,16	12	33	11	4	130		
CC8	3Q	tTHL	23,22,12,33,38,39,15,16	27	28	29	2	80		
BB8	3$\bar{Q}$	tTLH	23,22,12,33,38,39,15,16	27	28	9	4	130		
CC9	4Q	tTHL	23,22,12,33,27,28,15,16	38	39	40	2	80	nS	24,25

Parameter: Transition time, CL = 50 pF, VDD = 15V

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• TABLE XXI. Translation time (tTLH and tTHL) test - Continued.

Test cond	Parameter			Connect				Initial limits 25° ± 5°C ambient		Unit	Test fig
				A to pin	B to pin	C to pin	D to pin	Min	Max		
BB9	$4\overline{Q}$	Translation time, CL = 50 pF, VDD = 15V	tTLH	23,22,12,33,27,28,15,18	38	39	20	4	130	nS	24,25
CC10	5Q		tTHL	23,22,12,33,27,28,38,39	15	16	17	2	80		
BB10	$5\overline{Q}$	Translation time, CL = 50 pF, VDD = 15V	tTLH	23,22,12,33,27,28,38,39	15	16	36	4	130	nS	24,25

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* TABLE XXII. Propagation delay time (tPLH and tPHL) test.

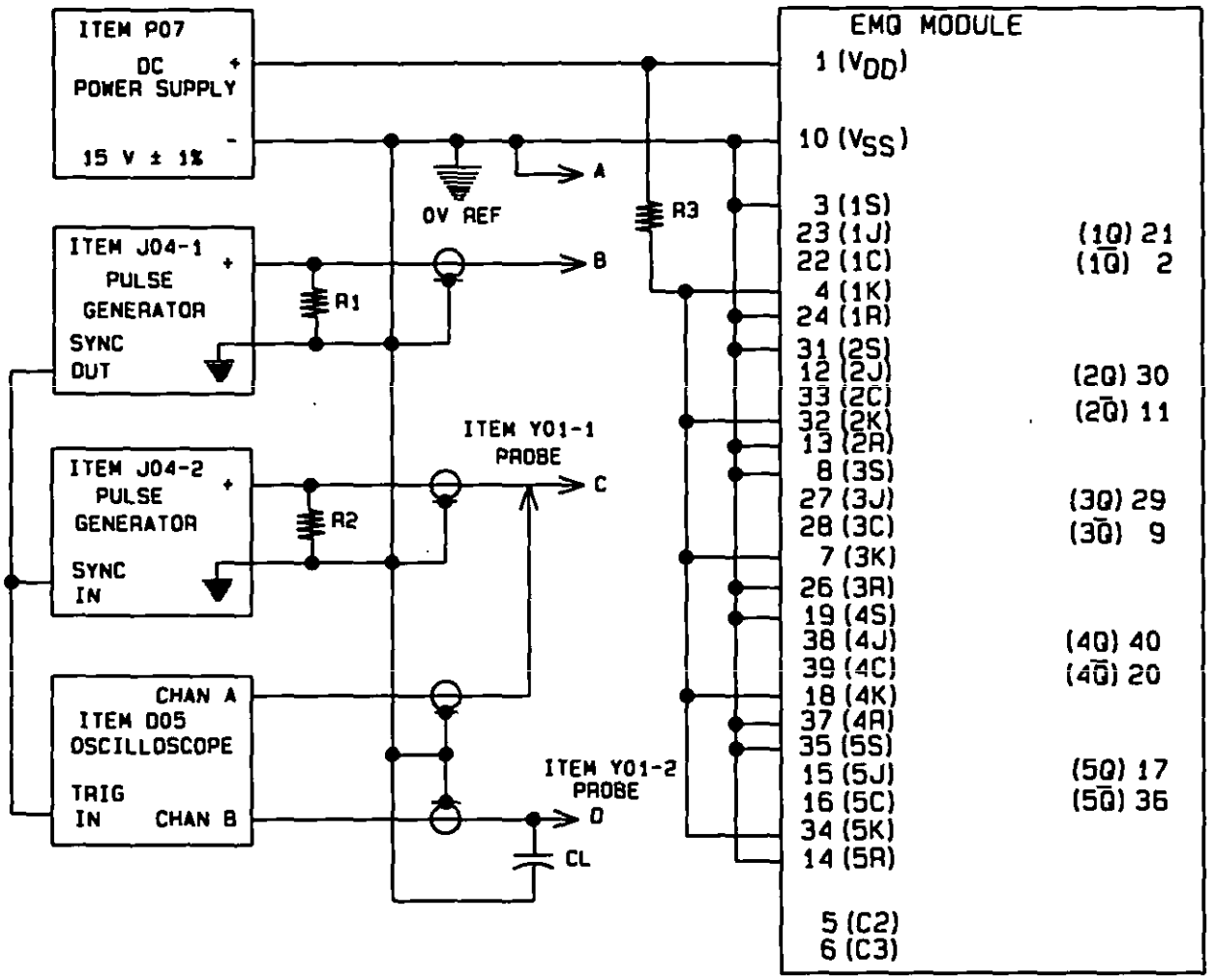
Test cond	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
DD1	1Q Propagation delay time, CL = 50 pF (clock to output), VDD = 15V	tPLH	12,33,27,28,38,39,15,16	23	22	21	3	100	nS	24,26
EE1	1$\bar{Q}$	tPHL	12,33,27,28,38,39,15,16	23	22	2				
DD2	2Q	tPLH	23,22,27,28,38,39,15,16	12	33	30				
EE2	2$\bar{Q}$	tPHL	23,22,27,28,38,39,15,16	12	33	11				
DD3	3Q	tPLH	23,22,12,33,38,39,15,16	27	28	29				
EE3	3$\bar{Q}$	tPHL	23,22,12,33,38,39,15,16	27	28	9				
DD4	4Q	tPLH	23,22,12,33,27,28,15,16	38	39	40				
EE4	4$\bar{Q}$	tPHL	23,22,12,33,27,28,15,16	38	39	20				
DD5	5Q	tPLH	23,22,12,33,27,28,38,39	15	16	17				
EE5	5$\bar{Q}$	tPHL	23,22,12,33,27,28,38,39	15	16	36				
EE6	1Q	tPHL	12,33,27,28,38,39,15,16	23	22	21				
DD6	1$\bar{Q}$	tPLH	12,33,27,28,38,39,15,16	23	22	2				
EE7	2Q	tPHL	23,22,27,38,28,39,15,16	12	33	30				
DD7	2$\bar{Q}$	tPLH	23,22,27,38,28,39,15,16	12	33	11				
EE8	3Q	tPHL	23,22,12,33,38,39,15,16	27	28	29				
DD8	3$\bar{Q}$	tPLH	23,22,12,33,38,39,15,16	27	28	9				
EE9	4Q Propagation delay time, CL = 50 pF (clock to output), VDD = 15V	tPHL	22,23,12,33,27,28,15,16	38	39	40	3	100	nS	24,26

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* TABLE XXII. Propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter			Connect				Initial limits 25° ± 5°C ambient		Unit	Test fig
				A to pin	B to pin	C to pin	D to pin	Min	Max		
DD9	$4\overline{Q}$	Propagation delay time, CL = 50 pF (clock to output), VDD = 15V	tPLH	22,23,12,33,27,28,15,18	38	39	20	3	100	nS	24,28
			tPHL	23,22,12,33,27,28,38,38	15	18	17	3	100	nS	24,28
EE10	5Q										
DD10	$5\overline{Q}$	Propagation delay time, CL = 50 pF (clock to output), VDD = 15V	tPLH	23,22,12,33,27,28,38,39	15	16	36				

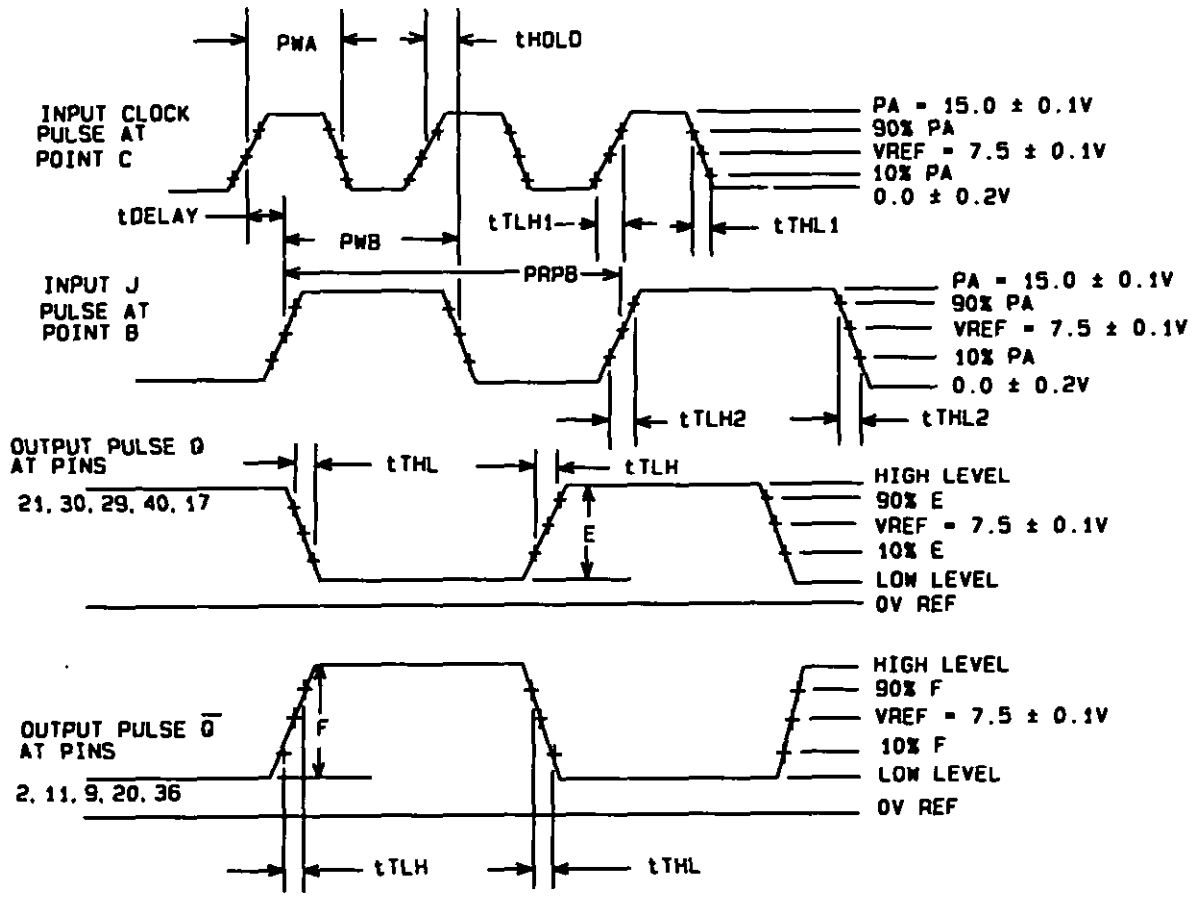
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NOTES:
1. R1 = R2 = 51.1 Ohms ± 5 percent, 2 W.
2. R3 = 1.5 kOhms ± 1 percent, 1/4 W.
3. CL = 50 ± 5 pF (including probe and parasitic capacitance).

FIGURE 24. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test circuit for tables XXI and XXII.

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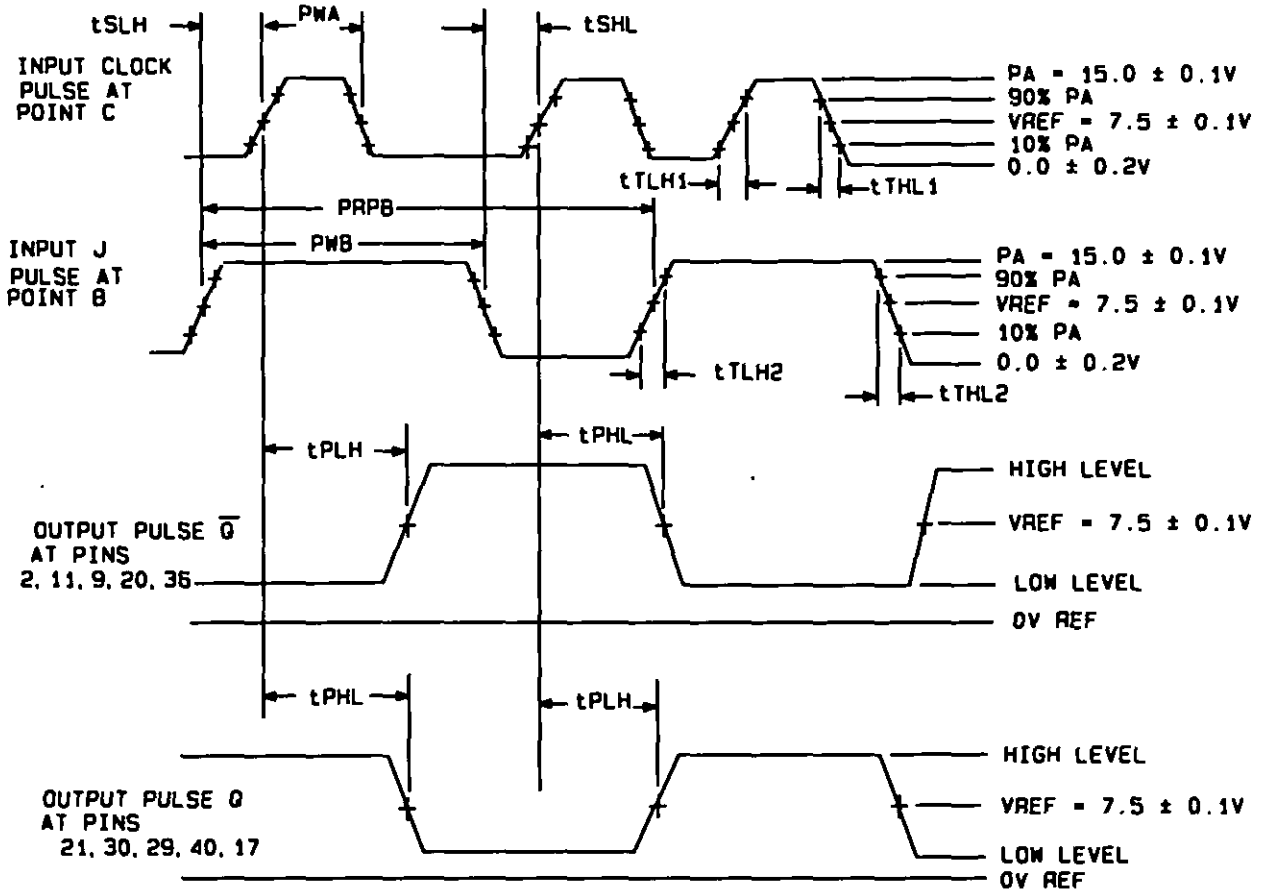


NOTES:

1. Input pulse parameters for item J04-2:
   Put item J04-2 in the double pulse mode.
   tTLH1 = tTHL1 = 10 ± 2 nS.
   PWA = 1.0 ± 0.1 µS.
2. Input pulse parameters for item J04-1:
   tTLH2 = tTHL2 = 10 ± 2 nS.
   PWB = 2.0 ± 0.2 µS.
   PRPB = 4.0 ± 0.4 µS.
3. tHOLD = 0 +2 nS, -0 nS.
   tDELAY = 1.0 ± 0.1 µS.
4. Verify that all outputs have a high level ≥ 13.5V and a low level ≤ 1.5V before making measurements.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
6. All voltages are referenced to ground.

FIGURE 25. Transition time (tTLH and tTHL) test circuit waveforms for table XXI.

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NOTES:

1. Input pulse parameters for item J04-2:
   Put item J04-2 in the double pulse mode.
   tTLH1 = tTHL1 = 10 ± 2 nS.
   PWA = 1.0 ± 0.1 µS.
2. Input pulse parameters for item J04-1:
   tTLH2 = tTHL2 = 10 ± 2 nS.
   PWB = 2.0 ± 0.2 µS.
   PRPB = 4.0 ± 0.4 µS.
3. tSLH = tSHL = 200 nS ± 5 percent.
4. Verify that all outputs have a high level ≥ 13.5V and a low level ≤ 1.5V before making measurements.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
6. All voltages are referenced to ground.

* FIGURE 26. Propagation delay time (tPLH and tPHL) test circuit waveforms for table XXII.

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* TABLE XXIII. Propagation delay time (tPLH and tPHL) test.

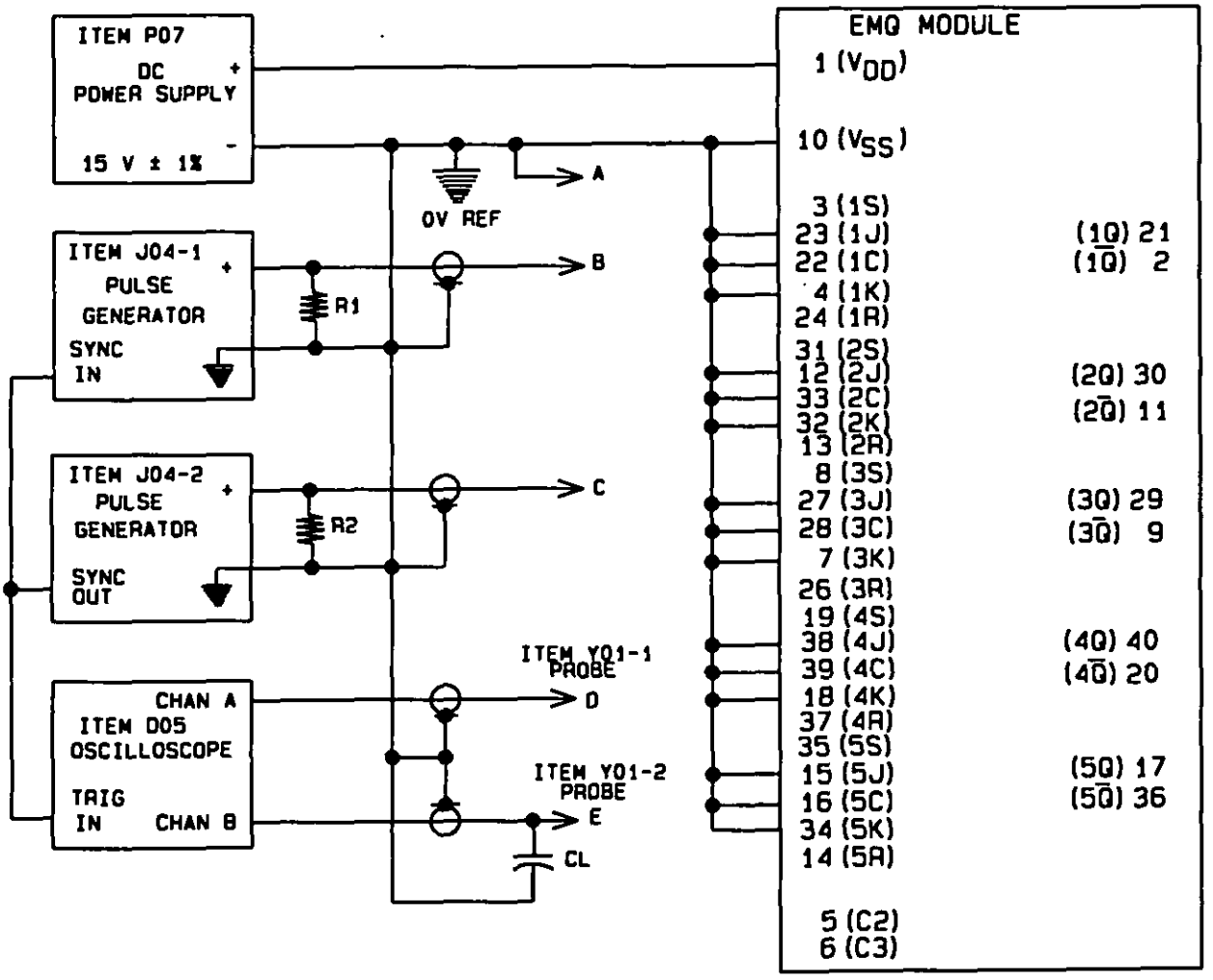
Test cond	Parameter		Connect: A to pin	B to pin	C to pin	D to pin	E to pin	Initial limits 25° ± 5°C ambient: Min	Max	Unit	Test fig
FF1	1Q Propagation delay time, CL = 50 pF (set to output), VDD = 15V	tPLH	31,13,8,26,19,37,35,14	3	24	B	21	3	100	nS	27,28
GG1	1$\bar{Q}$	tPHL	31,13,8,26,19,37,35,14	3	24		2				
FF2	2Q	tPLH	3,24,8,26,19,37,35,14	31	13		30				
GG2	2$\bar{Q}$	tPHL	3,24,8,26,19,37,35,14	31	13		11				
FF3	3Q	tPLH	3,24,31,13,19,37,35,14	8	26		29				
GG3	3$\bar{Q}$	tPHL	3,24,31,13,19,37,35,14	8	26		9				
FF4	4Q	tPLH	3,24,31,13,8,26,35,14	19	37		40				
GG4	4$\bar{Q}$	tPHL	3,24,31,13,8,26,35,14	19	37		20				
FF5	5Q	tPLH	3,24,31,13,8,26,19,37	35	14		17				
GG5	5$\bar{Q}$ Propagation delay time, CL = 50 pF (set to output), VDD = 15V	tPHL	3,24,31,13,8,26,19,37	35	14	B	38	3	100		
GG6	1Q Propagation delay time, CL = 50 pF (reset to output), VDD = 15V	tPHL	31,13,8,26,19,37,35,14	3	24	C	21	5	150		
FF6	1$\bar{Q}$	tPLH	31,13,8,26,19,37,35,14	3	24		2				
GG7	2Q	tPHL	3,24,8,26,19,37,35,14	31	13		30				
FF7	2$\bar{Q}$	tPLH	3,24,8,26,19,37,35,14	31	13		11				
GG8	3Q	tPHL	3,24,31,13,19,37,35,14	8	26		29				
FF8	3$\bar{Q}$ Propagation delay time, CL = 50 pF (reset to output), VDD = 15V	tPLH	3,24,31,13,19,37,35,14	8	26	C	9	5	150	nS	27,28

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* TABLE XXIII. Propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter			Connect						Initial limits 25° ± 5°C ambient		Unit	Test fig
				A to pin	B to pin	C to pin	D to pin	E to pin		Min	Max		
GG9	4Q	Propagation delay time, CL = 50 pF (reset to output), VDD = 15V	tPHL	3,24,31,13,8,26,35,14	19	37	C	40		5	150	nS	27,28
			tPLH		19	37		20					
FF8	$4\bar{Q}$		tPHL	3,24,31,13,8,26,35,14	35	14		17					
GG10	5Q		tPHL	3,24,31,13,8,26,19,37	35	14		17					
FF10	$5\bar{Q}$	Propagation delay time, CL = 50 pF (reset to output), VDD = 15V	tPLH	3,24,31,13,8,26,19,37	35	14	C	36		5	150	nS	27,28

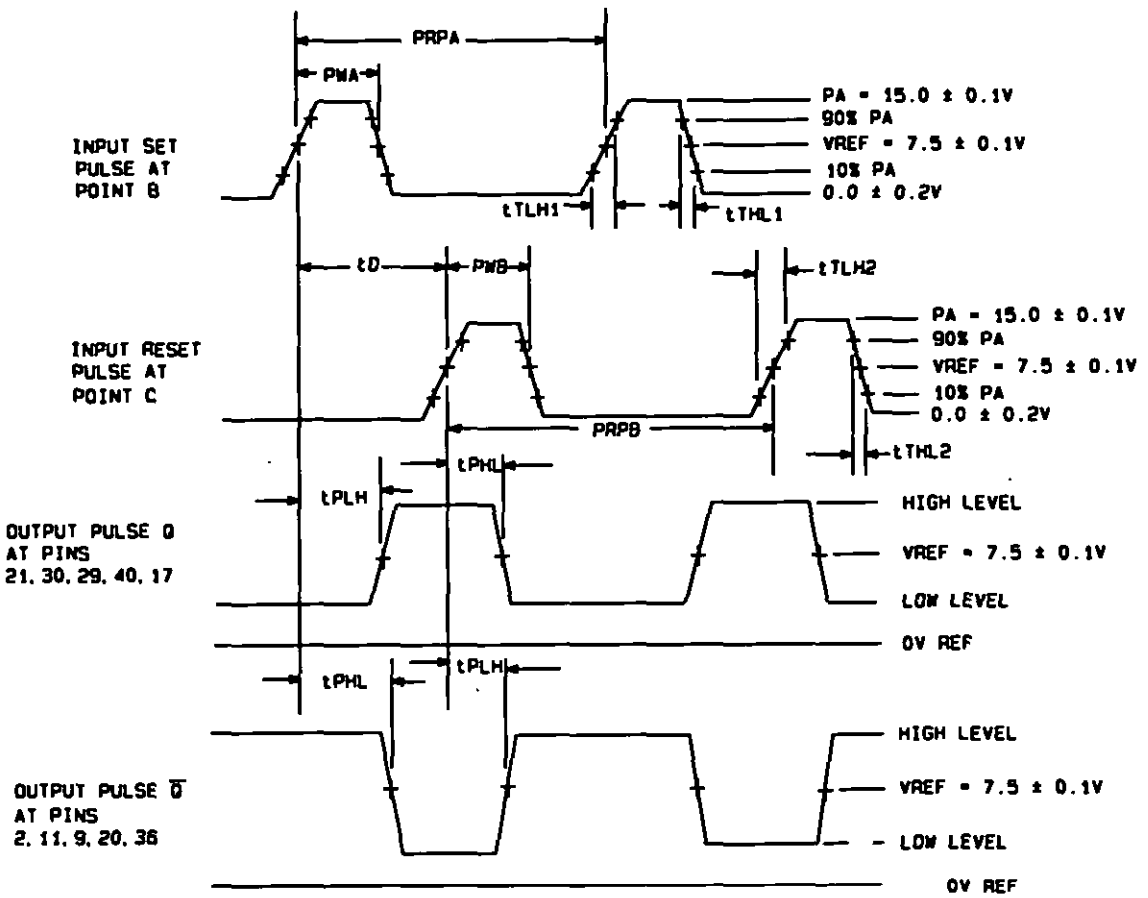
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NOTES:
1. R1 = R2 = 51.1 Ohms ± 5 percent, 2 W.
2. CL = 50 ± 5 pF (including probe and parasitic capacitance).

FIGURE 27. Propagation delay time (tPLH and tPHL) test circuit for table XXIII.

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NOTES:

1. Input pulse parameters for item J04-1:
   PRPA = 4.0 ± 0.4 µS.
   tTLH1 = tTHL1 = 10 ± 2 nS.
   PWA = 250 ± 20 nS.
2. Input pulse parameters for item J04-2:
   PRPB = 4.0 ± 0.4 µS.
   tTLH2 = tTHL2 = 10 ± 2 nS.
   PWB = 250 ± 20 nS.
3. tD = 2.0 ± 0.2 µS.
4. Verify that all outputs have a high level ≥ 13.5V and a low level ≤ 1.5V before making measurements.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
6. All voltages are referenced to ground.

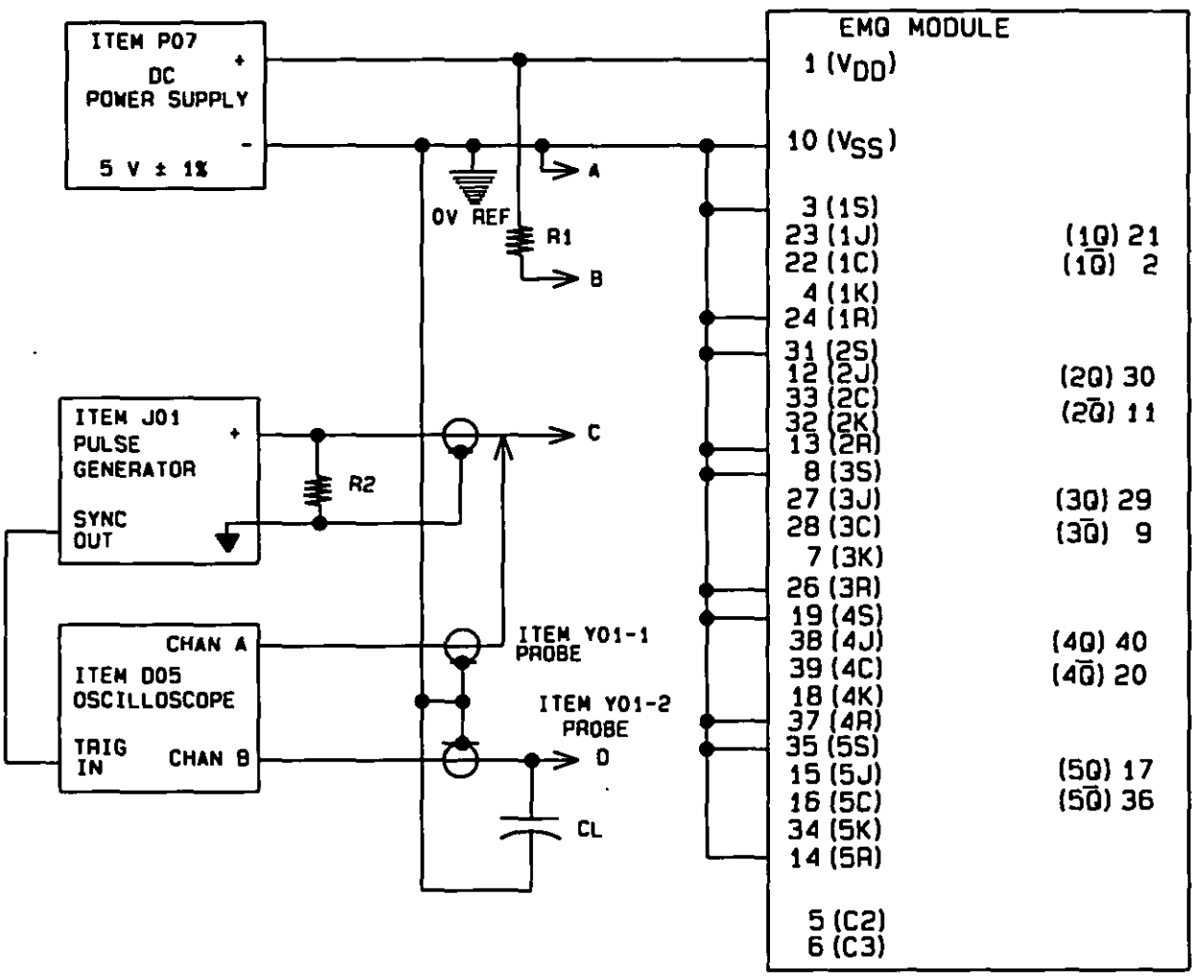
FIGURE 28. Propagation delay time (tPLH and tPHL) test circuit waveforms for table XXIII.

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TABLE XXIV. Maximum shift frequency (TA) test.

Test cond	Parameter	Connect A to pin	B to pin	C to pin	D to pin	Initial limits 25° ± 5°C ambient Min	Max	Unit	Test fig
HH1	1Q Maximum shift frequency (TA)	12,33,32,27,28,7, 38,39,18,15,16,34	23,4	22	21	1200	1465	nS	29,30
HH2	1$\bar{Q}$	12,33,32,27,28,7, 38,39,18,15,16,34	23,4	22	2				
HH3	2Q	23,22,4,27,28,7, 38,39,18,15,16,34	12,32	33	30				
HH4	2$\bar{Q}$	23,22,4,27,28,7, 38,39,18,15,16,34	12,32	33	11				
HH5	3Q	23,22,4,12,33,32, 38,39,18,15,16,34	27,7	28	29				
HH6	3$\bar{Q}$	23,22,4,12,33,32, 38,39,18,15,16,34	27,7	28	9				
HH7	4Q	23,22,4,12,33,32, 27,28,7,15,16,34	38,18	39	40				
HH8	4$\bar{Q}$	23,22,4,12,33,32, 27,28,7,15,16,34	38,18	39	20				
HH9	5Q	23,22,4,12,33,32, 27,28,7,38,39,18	15,34	16	17				
HH10	5$\bar{Q}$ Maximum shift frequency (TA)	23,22,4,12,33,32, 27,28,7,38,39,18	15,34	16	36	1200	1465	nS	29,30

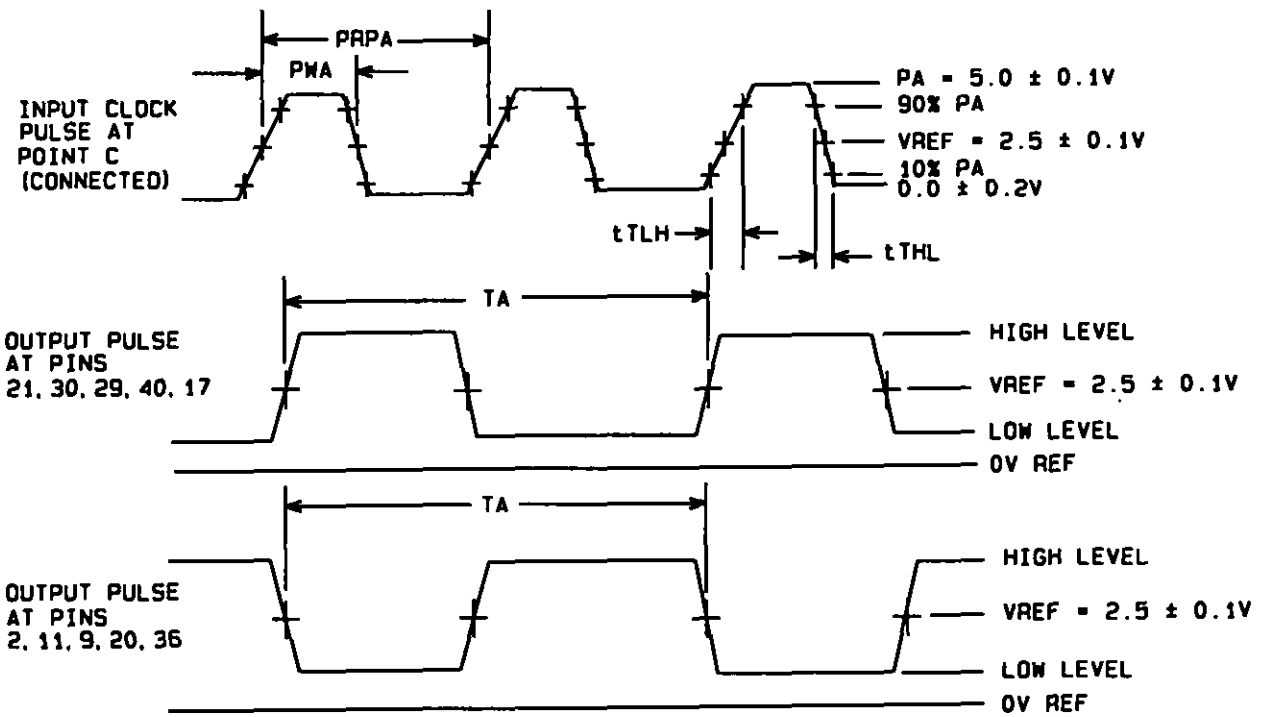
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NOTES:
1. R1 = R2 = 51.1 Ohms ± 5 percent, 2 W.
2. CL = 50 ± 5 pF (including probe and parasitic capacitance).

FIGURE 29. Maximum shift frequency (TA) test circuit for table XXIV.

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NOTES:
1. Input pulse parameters for item J01:
   tTLH = tTHL = 10 ± 2 nS.
   PWA = 330 +30, -0 nS.
   PRPA = 1.5 MHz ± 1 percent.
2. Verify that the output pulse high level is ≥ 4.6V and the low level is ≤ 0.4V before making the measurements.
3. The lead length of electrical conductors between resistor R2 and input pins items Y01-1 and Y01-2 to input and output pins, and between lead circuit and output pins shall be less than 1.00 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 30. Maximum shift frequency test circuit waveforms for table XXIV.

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TABLE XXV. Power supply filter capacitance.

Test cond	Parameter	Connect		Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
JJ	Power supply filter capacitance	Open	1	0.007	0.013	μF	31

TABLE XXVI. C2 and C3 capacitance test.

Test cond	Parameter	Connect		Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
KK1	C2 Capacitance	5	Open	176	264	pF	31
KK2	C3 Capacitance	6	Open	176	264	pF	31

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TABLE XXVII. Input capacitance test.

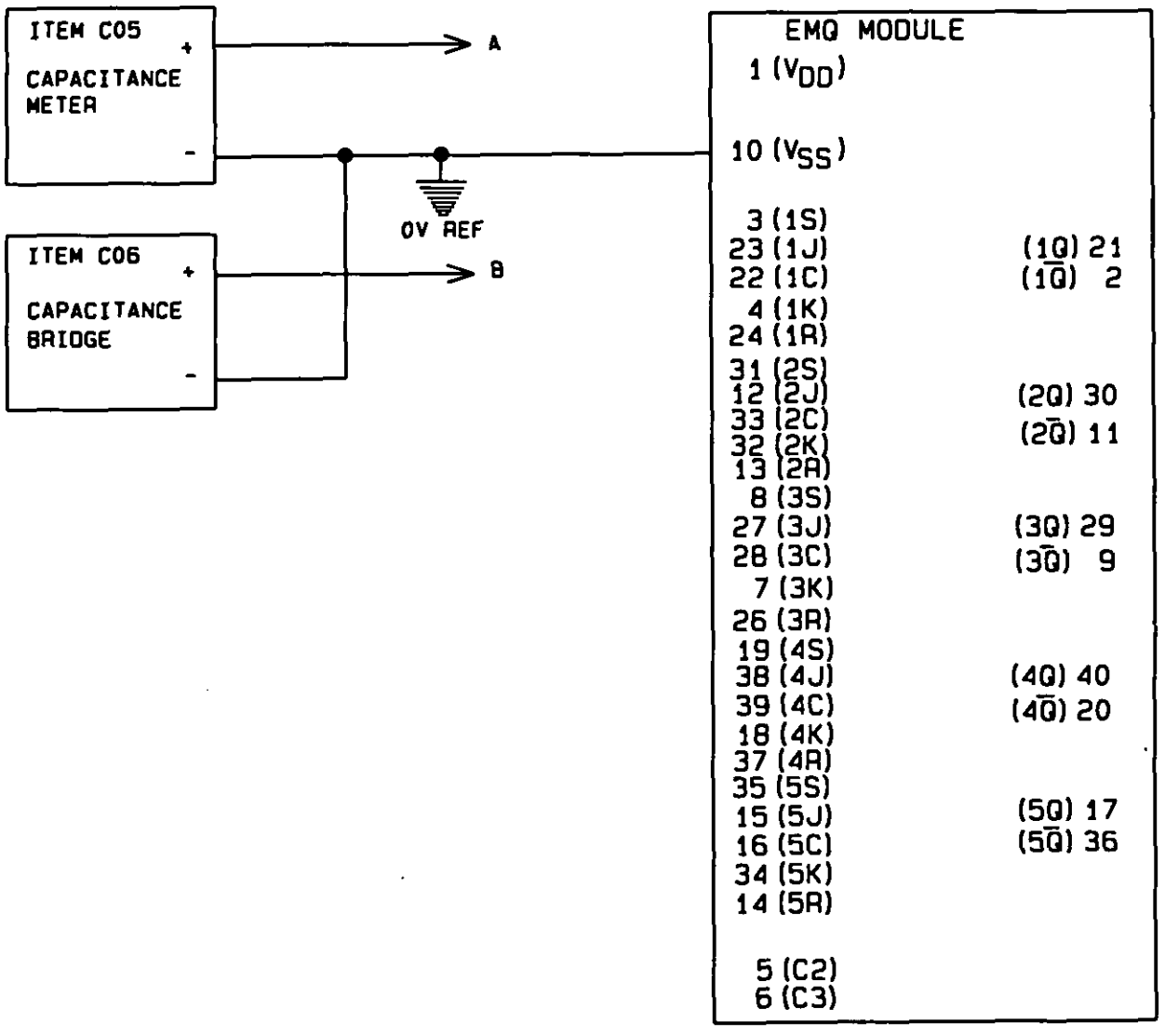
Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
LL1	1S Input capacitance	3	Open	2	15	pF	31
LL2	1J	23					
LL3	1C	22					
LL4	1K	4					
LL5	1R	24					
LL6	2S	31					
LL7	2J	12					
LL8	2C	33					
LL9	2K	32					
LL10	2R	13					
LL11	3S	8					
LL12	3J	27					
LL13	3C	28					
LL14	3K	7					
LL15	3R	26					
LL16	4S	19					
LL17	4J	38					
LL18	4C	39					
LL19	4K	18					
LL20	4R	37					
LL21	5S Input capacitance	35	Open	2	15	pF	31

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TABLE XXVII. Input capacitance test - Continued.

Test cond	Parameter	Connect		Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
LL22	5J Input capacitance	15	Open	2	15	pF	31
LL23	5C	16					
LL24	5K	34					
LL25	5R Input capacitance	14	Open	2	15	pF	31

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C06 setup:
   Frequency = 1 kHz.
   Vpk = 0.3 V maximum.

* FIGURE 31. Power supply filter, C2 and C3, and input capacitance test circuit for tables XXV, XXVI, and XXVII.

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3. DOCUMENT TITLE
Modules, Standard Electronic, Flip-Flop, JK, Digital, Key Code EMQ

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

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